

Imię i nazwisko: Paweł Borowiecki

Kompilacja jądra z wykorzystaniem localmodconfig

Na początku należy przejść do katalogu /usr/src/, za pomocą komendy cd i pobrać najnowsze źródła kernela, wykorzystując wget link_do_pobrania_kernela.

```
root@slack:~# cd /usr/src
root@slack:/usr/src# wget https://cdn.kernel.org/pub/linux/kernel/v7.x/linux-7.0.6.tar.xz
--2026-05-13 16:06:22-- https://cdn.kernel.org/pub/linux/kernel/v7.x/linux-7.0.6.tar.xz
Resolving cdn.kernel.org (cdn.kernel.org)... 151.101.5.176, 2a04:4e42:1::432
Connecting to cdn.kernel.org (cdn.kernel.org)|151.101.5.176|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 157158988 (150M) [application/x-xz]
Saving to: 'linux-7.0.6.tar.xz'

linux-7.0.6.tar.xz      100%[=====>] 149.88M  25.6MB/s   in 6.1s

2026-05-13 16:06:28 (24.6 MB/s) - 'linux-7.0.6.tar.xz' saved [157158988/157158988]

root@slack:/usr/src#
```

Następnie należy rozpakować plik archiwum ze źródłami kernela, za pomocą komendy tar z opcjami xvpf.

```
root@slack:/usr/src# tar -xvpf linux-7.0.6.tar.xz
```

```
linux-7.0.6/LICENSES/preferred/BSD-2-Clause
linux-7.0.6/LICENSES/preferred/BSD-3-Clause
linux-7.0.6/LICENSES/preferred/BSD-3-Clause-Clear
linux-7.0.6/LICENSES/preferred/GPL-2.0
linux-7.0.6/LICENSES/preferred/LGPL-2.0
linux-7.0.6/LICENSES/preferred/LGPL-2.1
linux-7.0.6/LICENSES/preferred/MIT
linux-7.0.6/MAINTAINERS
linux-7.0.6/Makefile
linux-7.0.6/README
linux-7.0.6/arch/
linux-7.0.6/arch/.gitignore
linux-7.0.6/arch/Kconfig
linux-7.0.6/arch/alpha/
linux-7.0.6/arch/alpha/Kbuild
linux-7.0.6/arch/alpha/Kconfig
linux-7.0.6/arch/alpha/Kconfig.debug
linux-7.0.6/arch/alpha/Makefile
linux-7.0.6/arch/alpha/boot/
linux-7.0.6/arch/alpha/boot/Makefile
linux-7.0.6/arch/alpha/boot/bootloader.lds
linux-7.0.6/arch/alpha/boot/bootp.c
linux-7.0.6/arch/alpha/boot/bootpz.c
linux-7.0.6/arch/alpha/boot/head.S
linux-7.0.6/arch/alpha/boot/main.c
linux-7.0.6/arch/alpha/boot/misc.c
linux-7.0.6/arch/alpha/boot/stdio.c
linux-7.0.6/arch/alpha/boot/tools/
linux-7.0.6/arch/alpha/boot/tools/mkbb.c
linux-7.0.6/arch/alpha/boot/tools/objstrip.c
linux-7.0.6/arch/alpha/configs/
linux-7.0.6/arch/alpha/configs/defconfig
linux-7.0.6/arch/alpha/include/
linux-7.0.6/arch/alpha/include/asm/
linux-7.0.6/arch/alpha/include/asm/Kbuild
linux-7.0.6/arch/alpha/include/asm/agg_backend.h
linux-7.0.6/arch/alpha/include/asm/asm-prototypes.h
linux-7.0.6/arch/alpha/include/asm/atomic.h
linux-7.0.6/arch/alpha/include/asm/barrier.h
linux-7.0.6/arch/alpha/include/asm/bitops.h
linux-7.0.6/arch/alpha/include/asm/bug.h
linux-7.0.6/arch/alpha/include/asm/cache.h
linux-7.0.6/arch/alpha/include/asm/cacheflush.h
linux-7.0.6/arch/alpha/include/asm/checksum.h
linux-7.0.6/arch/alpha/include/asm/cmpxchg.h
linux-7.0.6/arch/alpha/include/asm/compiler.h
linux-7.0.6/arch/alpha/include/asm/console.h
```

Potem trzeba przejść do folderu z rozpakowanym kernelu.

```
linux-7.0.6/01rc/kom/0110.c
linux-7.0.6/virt/kvm/vfio.h
linux-7.0.6/virt/lib/
linux-7.0.6/virt/lib/Kconfig
linux-7.0.6/virt/lib/Makefile
linux-7.0.6/virt/lib/irqbypass.c
root@slack:/usr/src# ls
linux@ linux-5.15.19/ linux-7.0.6/ linux-7.0.6.tar.xz
root@slack:/usr/src# cd linux-7.0.6
root@slack:/usr/src/linux-7.0.6#
```

Kolejny krok to skopiowanie konfiguracji starego jądra, za pomocą polecenia zcat.

```
root@slack:/usr/src/linux-7.0.6# zcat /proc/config.gz > .config
root@slack:/usr/src/linux-7.0.6# ls -al
total 1548
drwxrwxr-x 26 root root 4096 May 13 16:14 ./
drwxr-xr-x 4 root root 4096 May 13 16:08 ../
-rw-rw-r-- 1 root root 24291 May 11 08:21 .clang-format
-rw-rw-r-- 1 root root 374 May 11 08:21 .clippy.toml
-rw-rw-r-- 1 root root 59 May 11 08:21 .cocciconfig
-rw-rw-r-- 1 root root 239284 May 13 16:14 .config
-rw-rw-r-- 1 root root 566 May 11 08:21 .editorconfig
-rw-rw-r-- 1 root root 305 May 11 08:21 .get_maintainer.ignore
-rw-rw-r-- 1 root root 105 May 11 08:21 .gitattributes
-rw-rw-r-- 1 root root 2238 May 11 08:21 .gitignore
-rw-rw-r-- 1 root root 54512 May 11 08:21 .mailmap
-rw-rw-r-- 1 root root 66 May 11 08:21 .pylintrc
-rw-rw-r-- 1 root root 369 May 11 08:21 .rustfmt.toml
-rw-rw-r-- 1 root root 496 May 11 08:21 COPYING
-rw-rw-r-- 1 root root 107789 May 11 08:21 CREDITS
drwxrwxr-x 78 root root 4096 May 11 08:21 Documentation/
-rw-rw-r-- 1 root root 2871 May 11 08:21 Kbuild
-rw-rw-r-- 1 root root 582 May 11 08:21 Kconfig
drwxrwxr-x 6 root root 4096 May 11 08:21 LICENSES/
-rw-rw-r-- 1 root root 894639 May 11 08:21 MAINTAINERS
-rw-rw-r-- 1 root root 74251 May 11 08:21 Makefile
-rw-rw-r-- 1 root root 6043 May 11 08:21 README
drwxrwxr-x 23 root root 4096 May 11 08:21 arch/
drwxrwxr-x 3 root root 4096 May 11 08:21 block/
drwxrwxr-x 2 root root 4096 May 11 08:21 certs/
drwxrwxr-x 5 root root 4096 May 11 08:21 crypto/
drwxrwxr-x 147 root root 4096 May 11 08:21 drivers/
drwxrwxr-x 79 root root 4096 May 11 08:21 fs/
drwxrwxr-x 33 root root 4096 May 11 08:21 include/
drwxrwxr-x 2 root root 4096 May 11 08:21 init/
drwxrwxr-x 2 root root 4096 May 11 08:21 io_uring/
drwxrwxr-x 2 root root 4096 May 11 08:21 ipc/
drwxrwxr-x 24 root root 4096 May 11 08:21 kernel/
drwxrwxr-x 22 root root 12288 May 11 08:21 lib/
drwxrwxr-x 7 root root 4096 May 11 08:21 mm/
drwxrwxr-x 73 root root 4096 May 11 08:21 net/
drwxrwxr-x 11 root root 4096 May 11 08:21 rust/
drwxrwxr-x 47 root root 4096 May 11 08:21 samples/
drwxrwxr-x 23 root root 12288 May 11 08:21 scripts/
drwxrwxr-x 15 root root 4096 May 11 08:21 security/
drwxrwxr-x 27 root root 4096 May 11 08:21 sound/
drwxrwxr-x 47 root root 4096 May 11 08:21 tools/
drwxrwxr-x 4 root root 4096 May 11 08:21 usr/
drwxrwxr-x 4 root root 4096 May 11 08:21 virt/
root@slack:/usr/src/linux-7.0.6#
```

Następnie należy utworzyć plik z modułami, które są używane, za pomocą polecenia `make localmodconfig` i (tak jak na zajęciach) przytrzymać klawisz ENTER, akceptując domyślne dane.

```
root@slack:/usr/src/linux-7.0.6# make localmodconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/kconfig/conf.o
HOSTCC scripts/kconfig/confdata.o
HOSTCC scripts/kconfig/expr.o
LEX scripts/kconfig/lexer.lex.c
YACC scripts/kconfig/parser.tab.[ch]
HOSTCC scripts/kconfig/lexer.lex.o
HOSTCC scripts/kconfig/menu.o
HOSTCC scripts/kconfig/parser.tab.o
HOSTCC scripts/kconfig/preprocess.o
HOSTCC scripts/kconfig/symbol.o
HOSTCC scripts/kconfig/util.o
HOSTLD scripts/kconfig/conf
using config: '.config'
crc10dif_pclmul config not found!
crc32_pclmul config not found!
WARNING: FB_SYS_FILLRECT is required, but nothing in the
current config selects it.
WARNING: FB_SYSMEM_FOPS is required, but nothing in the
current config selects it.
WARNING: FB_SYS_COPYAREA is required, but nothing in the
current config selects it.
WARNING: FB_SYS_IMAGEBLIT is required, but nothing in the
current config selects it.
module fb_sys_fops did not have configs CONFIG_FB_SYSMEM_FOPS
```

```
Test functions located in the hexdump module at runtime (TEST_HEXDUMP) [N/m/y/?] n
Test kstrto*() family of functions at runtime (TEST_KSTRTOX) [N/m/y/?] n
Test bitmap_*() family of functions at runtime (TEST_BITMAP) [N/m/y/?] n
Test the XArray code at runtime (TEST_XARRAY) [N/m/y/?] n
Test the Maple Tree code at runtime or module load (TEST_MAPLE_TREE) [N/m/y/?] (NEW)
Perform selftest on resizable hash table (TEST_RHASHTABLE) [N/m/y/?] n
Perform selftest on IDA functions (TEST_IDA) [N/m/y/?] n
Test module loading with 'hello world' module (TEST_LKM) [N/m/?] n
Test module for compilation of bitops operations (TEST_BITOPS) [N/m/y/?] n
Test module for stress/performance analysis of vmalloc allocator (TEST_VMALLOCC) [N/m/y/?] n
Test BPF filter functionality (TEST_BPF) [N/m/?] n
Test find_bit functions (FIND_BIT_BENCHMARK) [N/m/y/?] n
Test firmware loading via userspace interface (TEST_FIRMWARE) [N/m/y/?] n
sysctl test driver (TEST_SYSCTL) [N/m/y/?] n
Compiler context-analysis warnings test (CONTEXT_ANALYSIS_TEST) [N/y/?] (NEW)
udelay test driver (TEST_UDELAY) [N/m/y/?] n
Test static keys (TEST_STATIC_KEYS) [N/m/?] n
kmod stress tester (TEST_KMOD) [N/m/?] n
module kallsyms find_symbol() test (TEST_KALLSYMS) [N/m/?] (NEW)
Test memcat_p() helper function (TEST_MEMCAT_P) [N/m/y/?] n
Test heap/page initialization (TEST_MEMINIT) [N/m/y/?] n
Test HMM (Heterogeneous Memory Management) (TEST_HMM) [N/m/y/?] n
Test freeing pages (TEST_FREE_PAGES) [N/m/y/?] n
Test floating point operations in kernel space (TEST_FPU) [N/m/y/?] n
Test clocksource watchdog in kernel space (TEST_CLOCKSOURCE_WATCHDOG) [N/m/y/?] n
Test module for correctness and stress of objpool (TEST_OBJPOOL) [N/m/?] (NEW)
#
# configuration written to .config
#
```

Potem trzeba sprawdzić, aktualnie załadowane moduły, poleceniem `lsmod`.

```
root@slack:/usr/src/linux-7.0.6# lsmod
```

```

nls_iso8859_1      16384  1
nls_cp437          20480  1
vfat              20480  1
fat               86016  1 vfat
intel_rapl_msr     20480  0
joydev            24576  0
intel_rapl_common  28672  1 intel_rapl_msr
crt10dif_pclmul   16384  1
crc32_pclmul      16384  0
vmwgfx            376832  1
ghash_clmulni_intel 16384  0
snd_intel8x0       45056  0
snd_ac97_codec     172032  1 snd_intel8x0
ohci_pci           16384  0
snd_pcm           139264  2 snd_intel8x0,snd_ac97_codec
evdev             28672  7
ttm               77824  1 vmwgfx
snd_timer         36864  1 snd_pcm
psmouse           167936  0
serio_raw         16384  0
snd               106496  4 snd_intel8x0,snd_timer,snd_ac97_codec,snd_pcm
drm_kms_helper    307200  1 vmwgfx
soundcore         16384  1 snd
ohci_hcd          49152  1 ohci_pci
ehci_pci          16384  0
i2c_piix4         24576  0
ac97_bus          16384  1 snd_ac97_codec
ehci_hcd          65536  1 ehci_pci
vboxguest         45056  0
drm               634880  4 vmwgfx,drm_kms_helper,ttm
intel_agp         20480  0
intel_gtt         24576  1 intel_agp
agpgart           53248  4 intel_agp,intel_gtt,ttm,drm
fb_sys_fops       16384  1 drm_kms_helper
syscopyarea       16384  1 drm_kms_helper
sysfillrect       16384  1 drm_kms_helper
ac                16384  0
sysimgblt         16384  1 drm_kms_helper
battery           24576  0
i2c_core          102400  4 drm_kms_helper,psmouse,i2c_piix4,drm
video             49152  0
e1000             131072  0
button            20480  0
loop              40960  0
ext4              921600  2
mbcache           16384  1 ext4
jbd2              163840  1 ext4

```

Później należy zbudować plik konfiguracyjny kernela z konfiguracją z poprzedniego kernela, za pomocą polecenia `make olddefconfig`.

```

root@slack:/usr/src/linux-7.0.6# make olddefconfig
#
# No change to .config
#

```

Kolejnym krokiem jest skompilowanie obrazu jądra, za pomocą polecenia `make bzImage`.

```

root@slack:/usr/src/linux-7.0.6# make bzImage

```

```
HOSTCC arch/x86/tools/relocs_32.o
HOSTCC arch/x86/tools/relocs_64.o
HOSTCC arch/x86/tools/relocs_common.o
HOSTLD arch/x86/tools/relocs
HOSTCC arch/x86/tools/udso2c
TOUCH include/generated/gcc-plugins.h
HOSTCC scripts/kallsyms
HOSTCC scripts/sorttable.o
HOSTCC scripts/elf-parse.o
HOSTLD scripts/sorttable
HOSTCC scripts/asn1_compiler
HOSTCC scripts/tracepoint-update.o
HOSTLD scripts/tracepoint-update
WRAP arch/x86/include/generated/uapi/asm/bpf_perf_event.h
WRAP arch/x86/include/generated/uapi/asm/errno.h
WRAP arch/x86/include/generated/uapi/asm/fcntl.h
WRAP arch/x86/include/generated/uapi/asm/ioctl.h
WRAP arch/x86/include/generated/uapi/asm/ioctls.h
WRAP arch/x86/include/generated/uapi/asm/ipcbuf.h
WRAP arch/x86/include/generated/uapi/asm/param.h
WRAP arch/x86/include/generated/uapi/asm/poll.h
WRAP arch/x86/include/generated/uapi/asm/resource.h
WRAP arch/x86/include/generated/uapi/asm/socket.h
WRAP arch/x86/include/generated/uapi/asm/sockios.h
WRAP arch/x86/include/generated/uapi/asm/termbits.h
WRAP arch/x86/include/generated/uapi/asm/termios.h
WRAP arch/x86/include/generated/uapi/asm/types.h
WRAP arch/x86/include/generated/asm/early_ioremap.h
WRAP arch/x86/include/generated/asm/fprobe.h
WRAP arch/x86/include/generated/asm/mcs_spinlock.h
WRAP arch/x86/include/generated/asm/mmzone.h
WRAP arch/x86/include/generated/asm/irq_regs.h
WRAP arch/x86/include/generated/asm/kmap_size.h
WRAP arch/x86/include/generated/asm/local64.h
WRAP arch/x86/include/generated/asm/mmiowb.h
WRAP arch/x86/include/generated/asm/module.lds.h
WRAP arch/x86/include/generated/asm/rwonce.h
UPD arch/x86/include/generated/asm/cpufeaturemasks.h
GEN arch/x86/include/generated/asm/orc_hash.h
UPD include/config/kernel.release
UPD include/generated/uapi/linux/version.h
UPD include/generated/utsrelease.h
UPD include/generated/compile.h
CC scripts/mod/empty.o
HOSTCC scripts/mod/mk_elfconfig
MKELF scripts/mod/elfconfig.h
HOSTCC scripts/mod/modpost.o
```

```
CC      arch/x86/coco/sev/noinstr.o
CC      arch/x86/coco/sev/vc-handle.o
CC      arch/x86/coco/sev/svsm.o
AR      arch/x86/coco/sev/built-in.a
CC      arch/x86/coco/core.o
AR      arch/x86/coco/built-in.a
LDS      arch/x86/entry/udso/udso64/udso64.lds
AS      arch/x86/entry/udso/udso64/note.o
CC      arch/x86/entry/udso/udso64/vclock_gettime.o
CC      arch/x86/entry/udso/udso64/vgetcpu.o
CC      arch/x86/entry/udso/udso64/vgetrandom.o
AS      arch/x86/entry/udso/udso64/vgetrandom-chacha.o
AS      arch/x86/entry/udso/udso64/vsgx.o
VDSO    arch/x86/entry/udso/udso64/udso64.so.dbg
OBJCOPY arch/x86/entry/udso/udso64/udso64.so
VDSO2C  arch/x86/entry/udso/udso64/udso64-image.c
CC      arch/x86/entry/udso/udso64/udso64-image.o
AR      arch/x86/entry/udso/udso64/built-in.a
LDS      arch/x86/entry/udso/udso32/udso32.lds
AS      arch/x86/entry/udso/udso32/note.o
CC      arch/x86/entry/udso/udso32/vclock_gettime.o
CC      arch/x86/entry/udso/udso32/vgetcpu.o
AS      arch/x86/entry/udso/udso32/system_call.o
AS      arch/x86/entry/udso/udso32/sigreturn.o
VDSO    arch/x86/entry/udso/udso32/udso32.so.dbg
OBJCOPY arch/x86/entry/udso/udso32/udso32.so
VDSO2C  arch/x86/entry/udso/udso32/udso32-image.c
CC      arch/x86/entry/udso/udso32/udso32-image.o
AR      arch/x86/entry/udso/udso32/built-in.a
CC      arch/x86/entry/udso/uma.o
CC      arch/x86/entry/udso/extable.o
CC      arch/x86/entry/udso/udso32-setup.o
AR      arch/x86/entry/udso/built-in.a
CC      arch/x86/entry/vsyscall/vsyscall_64.o
AS      arch/x86/entry/vsyscall/vsyscall_emu_64.o
AR      arch/x86/entry/vsyscall/built-in.a
AS      arch/x86/entry/entry.o
AS      arch/x86/entry/entry_64.o
CC      arch/x86/entry/syscall_64.o
AS      arch/x86/entry/thunk.o
AS      arch/x86/entry/entry_64_compat.o
CC      arch/x86/entry/syscall_32.o
AR      arch/x86/entry/built-in.a
CC      arch/x86/events/amd/core.o
CC      arch/x86/events/amd/lbr.o
CC      arch/x86/events/amd/ibs.o
CC      arch/x86/events/amd/uncore.o
```

```
CC arch/x86/kernel/irq_work.o
CC arch/x86/kernel/probe_roms.o
CC arch/x86/kernel/sys_ia32.o
CC arch/x86/kernel/signal_32.o
CC arch/x86/kernel/sys_x86_64.o
CC arch/x86/kernel/espfix_64.o
CC arch/x86/kernel/ksysfs.o
CC arch/x86/kernel/bootflag.o
CC arch/x86/kernel/e820.o
CC arch/x86/kernel/pci-dma.o
CC arch/x86/kernel/quirks.o
CC arch/x86/kernel/kdebugfs.o
CC arch/x86/kernel/alternative.o
CC arch/x86/kernel/i8253.o
CC arch/x86/kernel/hw_breakpoint.o
CC arch/x86/kernel/tsc.o
CC arch/x86/kernel/tsc_msr.o
CC arch/x86/kernel/io_delay.o
CC arch/x86/kernel/rtc.o
CC arch/x86/kernel/resource.o
AS arch/x86/kernel/irqflags.o
CC arch/x86/kernel/static_call.o
CC arch/x86/kernel/process.o
CC arch/x86/kernel/ptrace.o
CC arch/x86/kernel/tls.o
CC arch/x86/kernel/step.o
CC arch/x86/kernel/i8237.o
CC arch/x86/kernel/stacktrace.o
CC arch/x86/kernel/reboot.o
CC arch/x86/kernel/msr.o
CC arch/x86/kernel/cpuid.o
CC arch/x86/kernel/early-quirks.o
CC arch/x86/kernel/smp.o
CC arch/x86/kernel/smpboot.o
CC arch/x86/kernel/tsc_sync.o
CC arch/x86/kernel/setup_percpu.o
CC arch/x86/kernel/mpparse.o
CC arch/x86/kernel/ftrace.o
AS arch/x86/kernel/ftrace_64.o
CC arch/x86/kernel/trace_clock.o
CC arch/x86/kernel/trace.o
CC arch/x86/kernel/vmcore_info_64.o
CC arch/x86/kernel/machine_kexec_64.o
AS arch/x86/kernel/relocate_kernel_64.o
CC arch/x86/kernel/kexec-bzimage64.o
CC arch/x86/kernel/crash_dump_64.o
CC arch/x86/kernel/crash.o
```

```
CC      drivers/firmware/efi/libstub/find.o
STUBCPY drivers/firmware/efi/libstub/find.stub.o
CC      drivers/firmware/efi/libstub/gop.o
STUBCPY drivers/firmware/efi/libstub/gop.stub.o
CC      drivers/firmware/efi/libstub/lib-cmdline.o
STUBCPY drivers/firmware/efi/libstub/lib-cmdline.stub.o
CC      drivers/firmware/efi/libstub/lib-ctype.o
STUBCPY drivers/firmware/efi/libstub/lib-ctype.stub.o
CC      drivers/firmware/efi/libstub/mem.o
STUBCPY drivers/firmware/efi/libstub/mem.stub.o
CC      drivers/firmware/efi/libstub/pci.o
STUBCPY drivers/firmware/efi/libstub/pci.stub.o
CC      drivers/firmware/efi/libstub/printk.o
STUBCPY drivers/firmware/efi/libstub/printk.stub.o
CC      drivers/firmware/efi/libstub/random.o
STUBCPY drivers/firmware/efi/libstub/random.stub.o
CC      drivers/firmware/efi/libstub/randomalloc.o
STUBCPY drivers/firmware/efi/libstub/randomalloc.stub.o
CC      drivers/firmware/efi/libstub/relocate.o
STUBCPY drivers/firmware/efi/libstub/relocate.stub.o
CC      drivers/firmware/efi/libstub/secureboot.o
STUBCPY drivers/firmware/efi/libstub/secureboot.stub.o
CC      drivers/firmware/efi/libstub/skip_spaces.o
STUBCPY drivers/firmware/efi/libstub/skip_spaces.stub.o
CC      drivers/firmware/efi/libstub/smbios.o
STUBCPY drivers/firmware/efi/libstub/smbios.stub.o
CC      drivers/firmware/efi/libstub/tpm.o
STUBCPY drivers/firmware/efi/libstub/tpm.stub.o
CC      drivers/firmware/efi/libstub/unaccepted_memory.o
STUBCPY drivers/firmware/efi/libstub/unaccepted_memory.stub.o
CC      drivers/firmware/efi/libstub/vsprintf.o
STUBCPY drivers/firmware/efi/libstub/vsprintf.stub.o
CC      drivers/firmware/efi/libstub/x86-5lvl.o
STUBCPY drivers/firmware/efi/libstub/x86-5lvl.stub.o
CC      drivers/firmware/efi/libstub/x86-stub.o
STUBCPY drivers/firmware/efi/libstub/x86-stub.stub.o
AR      drivers/firmware/efi/libstub/lib.a
CC      drivers/firmware/efi/efi-bgrt.o
CC      drivers/firmware/efi/efi.o
CC      drivers/firmware/efi/vars.o
CC      drivers/firmware/efi/reboot.o
CC      drivers/firmware/efi/memattr.o
CC      drivers/firmware/efi/tpm.o
CC      drivers/firmware/efi/memmap.o
CC      drivers/firmware/efi/esrt.o
CC      drivers/firmware/efi/efi-pstore.o
CC      drivers/firmware/efi/cper.o
```



```

AS      arch/x86/boot/compressed/head_64.o
VOFFSET arch/x86/boot/compressed/./voffset.h
CC      arch/x86/boot/compressed/misc.o
CC      arch/x86/boot/compressed/string.o
CC      arch/x86/boot/compressed/cmdline.o
CC      arch/x86/boot/compressed/error.o
OBJCOPY arch/x86/boot/compressed/vmlinux.bin
RELOCS  arch/x86/boot/compressed/vmlinux.relocs
LZMA     arch/x86/boot/compressed/vmlinux.bin.lzma
HOSTCC   arch/x86/boot/compressed/mkpiggy
MKPIGGY  arch/x86/boot/compressed/piggy.S
AS      arch/x86/boot/compressed/piggy.o
CC      arch/x86/boot/compressed/cpuflags.o
CC      arch/x86/boot/compressed/early_serial_console.o
CC      arch/x86/boot/compressed/kaslr.o
CC      arch/x86/boot/compressed/ident_map_64.o
CC      arch/x86/boot/compressed/idt_64.o
AS      arch/x86/boot/compressed/idt_handlers_64.o
AS      arch/x86/boot/compressed/mem_encrypt.o
CC      arch/x86/boot/compressed/pgtable_64.o
CC      arch/x86/boot/compressed/sev.o
CC      arch/x86/boot/compressed/sev-handle-vc.o
CC      arch/x86/boot/compressed/acpi.o
CC      arch/x86/boot/compressed/mem.o
CC      arch/x86/boot/compressed/efi.o
LD       arch/x86/boot/compressed/vmlinux
ZOFFSET  arch/x86/boot/zoffset.h
AS      arch/x86/boot/header.o
CC      arch/x86/boot/main.o
CC      arch/x86/boot/memory.o
CC      arch/x86/boot/pm.o
AS      arch/x86/boot/pm_jump.o
CC      arch/x86/boot/printf.o
CC      arch/x86/boot/regs.o
CC      arch/x86/boot/string.o
CC      arch/x86/boot/tty.o
CC      arch/x86/boot/video.o
CC      arch/x86/boot/video-mode.o
CC      arch/x86/boot/version.o
CC      arch/x86/boot/video-vga.o
CC      arch/x86/boot/video-vesa.o
CC      arch/x86/boot/video-bios.o
LD       arch/x86/boot/setup.elf
OBJCOPY  arch/x86/boot/setup.bin
OBJCOPY  arch/x86/boot/vmlinux.bin
BUILD    arch/x86/boot/bzImage
Kernel: arch/x86/boot/bzImage is ready (#1)
root@slack:/usr/src/linux-7.0.6#

```

Następnie należy zbudować moduły jądra, za pomocą polecenia `make modules`.

```
root@slack:/usr/src/linux-7.0.6# make modules
CALL      scripts/checksyscalls.sh
DESCEND   objtool
INSTALL   libsubcmd_headers
AS [M]    arch/x86/crypto/ghash-clmulni-intel_asm.o
CC [M]    arch/x86/crypto/ghash-clmulni-intel_glue.o
LD [M]    arch/x86/crypto/ghash-clmulni-intel.o
CC [M]    fs/nls/nls_cp437.o
CC [M]    fs/nls/nls_iso8859-1.o
CC [M]    fs/ext4/balloc.o
CC [M]    fs/ext4/bitmap.o
CC [M]    fs/ext4/block_validity.o
CC [M]    fs/ext4/dir.o
CC [M]    fs/ext4/ext4_jbd2.o
CC [M]    fs/ext4/extents.o
CC [M]    fs/ext4/extents_status.o
CC [M]    fs/ext4/file.o
CC [M]    fs/ext4/fsmap.o
CC [M]    fs/ext4/fsync.o
CC [M]    fs/ext4/hash.o
CC [M]    fs/ext4/ialloc.o
CC [M]    fs/ext4/indirect.o
CC [M]    fs/ext4/inline.o
CC [M]    fs/ext4/inode.o
```

```
LD [M] fs/ext4/ext4.o
CC [M] fs/ext2/balloc.o
CC [M] fs/ext2/dir.o
CC [M] fs/ext2/file.o
CC [M] fs/ext2/ialloc.o
CC [M] fs/ext2/inode.o
CC [M] fs/ext2/ioctl.o
CC [M] fs/ext2/namei.o
CC [M] fs/ext2/super.o
CC [M] fs/ext2/symlink.o
CC [M] fs/ext2/trace.o
CC [M] fs/ext2/xattr.o
CC [M] fs/ext2/xattr_user.o
CC [M] fs/ext2/xattr_trusted.o
CC [M] fs/ext2/acl.o
CC [M] fs/ext2/xattr_security.o
LD [M] fs/ext2/ext2.o
CC [M] fs/jbd2/transaction.o
CC [M] fs/jbd2/commit.o
CC [M] fs/jbd2/recovery.o
CC [M] fs/jbd2/checkpoint.o
CC [M] fs/jbd2/revoke.o
CC [M] fs/jbd2/journal.o
LD [M] fs/jbd2/jbd2.o
CC [M] fs/fat/cache.o
CC [M] fs/fat/dir.o
CC [M] fs/fat/fatent.o
CC [M] fs/fat/file.o
CC [M] fs/fat/inode.o
CC [M] fs/fat/misc.o
CC [M] fs/fat/nfs.o
LD [M] fs/fat/fat.o
CC [M] fs/fat/namei_vfat.o
LD [M] fs/fat/vfat.o
CC [M] fs/efivarfs/inode.o
CC [M] fs/efivarfs/file.o
CC [M] fs/efivarfs/super.o
CC [M] fs/efivarfs/vars.o
LD [M] fs/efivarfs/efivarfs.o
CC [M] fs/mbcache.o
CC [M] lib/crc/crc16.o
CC [M] drivers/acpi/ac.o
CC [M] drivers/acpi/button.o
CC [M] drivers/acpi/acpi_video.o
CC [M] drivers/acpi/video_detect.o
LD [M] drivers/acpi/video.o
CC [M] drivers/acpi/battery.o
```

```
CC [M] fs/efivarfs/inode.o
CC [M] fs/efivarfs/file.o
CC [M] fs/efivarfs/super.o
CC [M] fs/efivarfs/vars.o
LD [M] fs/efivarfs/efivarfs.o
CC [M] fs/mbcache.o
CC [M] lib/crc/crc16.o
CC [M] drivers/acpi/ac.o
CC [M] drivers/acpi/button.o
CC [M] drivers/acpi/acpi_video.o
CC [M] drivers/acpi/video_detect.o
LD [M] drivers/acpi/video.o
CC [M] drivers/acpi/battery.o
CC [M] drivers/char/agp/backend.o
CC [M] drivers/char/agp/generic.o
CC [M] drivers/char/agp/isoch.o
LD [M] drivers/char/agp/agpgart.o
CC [M] drivers/char/agp/intel-agp.o
CC [M] drivers/char/agp/intel-gtt.o
CC [M] drivers/gpu/drm/clients/drm_client_setup.o
CC [M] drivers/gpu/drm/clients/drm_fbdev_client.o
LD [M] drivers/gpu/drm/clients/drm_client_lib.o
CC [M] drivers/gpu/drm/ttm/ttm_tt.o
CC [M] drivers/gpu/drm/ttm/ttm_bo.o
CC [M] drivers/gpu/drm/ttm/ttm_bo_util.o
CC [M] drivers/gpu/drm/ttm/ttm_bo_vm.o
CC [M] drivers/gpu/drm/ttm/ttm_module.o
CC [M] drivers/gpu/drm/ttm/ttm_execbuf_util.o
CC [M] drivers/gpu/drm/ttm/ttm_range_manager.o
CC [M] drivers/gpu/drm/ttm/ttm_resource.o
CC [M] drivers/gpu/drm/ttm/ttm_pool.o
CC [M] drivers/gpu/drm/ttm/ttm_device.o
CC [M] drivers/gpu/drm/ttm/ttm_sys_manager.o
CC [M] drivers/gpu/drm/ttm/ttm_backup.o
CC [M] drivers/gpu/drm/ttm/ttm_agp_backend.o
LD [M] drivers/gpu/drm/ttm/ttm.o
CC [M] drivers/gpu/drm/umwgfx/umwgfx_execbuf.o
CC [M] drivers/gpu/drm/umwgfx/umwgfx_gmr.o
CC [M] drivers/gpu/drm/umwgfx/umwgfx_kms.o
CC [M] drivers/gpu/drm/umwgfx/umwgfx_drv.o
CC [M] drivers/gpu/drm/umwgfx/umwgfx_ioctl.o
CC [M] drivers/gpu/drm/umwgfx/umwgfx_resource.o
CC [M] drivers/gpu/drm/umwgfx/umwgfx_ttm_buffer.o
CC [M] drivers/gpu/drm/umwgfx/umwgfx_cmd.o
CC [M] drivers/gpu/drm/umwgfx/umwgfx_irq.o
CC [M] drivers/gpu/drm/umwgfx/umwgfx_ldu.o
CC [M] drivers/gpu/drm/umwgfx/umwgfx_overlay.o
```

```

LD [M] drivers/i2c/i2c-smbus.ko
CC [M] drivers/platform/x86/dell/dell-wmi-ddv.mod.o
LD [M] drivers/platform/x86/dell/dell-wmi-ddv.ko
CC [M] drivers/platform/wmi/wmi.mod.o
LD [M] drivers/platform/wmi/wmi.ko
CC [M] drivers/virt/vboxguest/vboxguest.mod.o
LD [M] drivers/virt/vboxguest/vboxguest.ko
CC [M] drivers/virt/coco/guest/tsm_report.mod.o
LD [M] drivers/virt/coco/guest/tsm_report.ko
CC [M] drivers/virt/coco/sev-guest/sev-guest.mod.o
LD [M] drivers/virt/coco/sev-guest/sev-guest.ko
CC [M] drivers/powercap/intel_rapl_common.mod.o
LD [M] drivers/powercap/intel_rapl_common.ko
CC [M] drivers/powercap/intel_rapl_msr.mod.o
LD [M] drivers/powercap/intel_rapl_msr.ko
CC [M] sound/soundcore.mod.o
LD [M] sound/soundcore.ko
CC [M] sound/core/snd.mod.o
LD [M] sound/core/snd.ko
CC [M] sound/core/snd-timer.mod.o
LD [M] sound/core/snd-timer.ko
CC [M] sound/core/snd-pcm.mod.o
LD [M] sound/core/snd-pcm.ko
CC [M] sound/pci/snd-intel8x0.mod.o
LD [M] sound/pci/snd-intel8x0.ko
CC [M] sound/pci/ac97/snd-ac97-codec.mod.o
LD [M] sound/pci/ac97/snd-ac97-codec.ko
CC [M] sound/ac97_bus.mod.o
LD [M] sound/ac97_bus.ko
CC [M] net/802/psnap.mod.o
LD [M] net/802/psnap.ko
CC [M] net/802/stp.mod.o
LD [M] net/802/stp.ko
CC [M] net/802/garp.mod.o
LD [M] net/802/garp.ko
CC [M] net/802/mrp.mod.o
LD [M] net/802/mrp.ko
CC [M] net/ipv6/ipv6.mod.o
LD [M] net/ipv6/ipv6.ko
CC [M] net/8021q/8021q.mod.o
LD [M] net/8021q/8021q.ko
CC [M] net/wireless/cfg80211.mod.o
LD [M] net/wireless/cfg80211.ko
CC [M] net/llc/llc.mod.o
LD [M] net/llc/llc.ko
CC [M] net/rfkill/rfkill.mod.o
LD [M] net/rfkill/rfkill.ko
root@slack:/usr/src/linux-7.0.6#

```

Kolejnym krokiem jest instalacja modułów, za pomocą komendy `make modules_install`.

```
root@slack:/usr/src/linux-7.0.6# make modules install
```

```

INSTALL /lib/modules/7.0.6/kernel/drivers/acpi/battery.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/char/agp/agpgart.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/char/agp/intel-agp.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/char/agp/intel-gtt.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/gpu/drm/clients/drm_client_lib.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/gpu/drm/drm.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/gpu/drm/drm_ttm_helper.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/gpu/drm/drm_kms_helper.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/gpu/drm/ttm/ttm.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/gpu/drm/umwgfx/umwgfx.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/block/loop.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/net/ethernet/intel/e1000/e1000.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/usb/host/ehci-hcd.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/usb/host/ehci-pci.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/usb/host/ohci-hcd.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/usb/host/ohci-pci.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/input/serio/serio_raw.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/input/mouse/psmouse.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/input/joydev.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/input/evdev.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/i2c/busses/i2c-piix4.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/i2c/i2c-core.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/i2c/i2c-smbus.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/platform/x86/dell/dell-wmi-ddv.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/platform/wmi/wmi.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/virt/uboxguest/uboxguest.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/virt/coco/guest/tsm_report.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/virt/coco/sev-guest/sev-guest.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/powercap/intel_rapl_common.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/powercap/intel_rapl_msr.ko
INSTALL /lib/modules/7.0.6/kernel/sound/soundcore.ko
INSTALL /lib/modules/7.0.6/kernel/sound/core/snd.ko
INSTALL /lib/modules/7.0.6/kernel/sound/core/snd-timer.ko
INSTALL /lib/modules/7.0.6/kernel/sound/core/snd-pcm.ko
INSTALL /lib/modules/7.0.6/kernel/sound/pci/snd-intel8x0.ko
INSTALL /lib/modules/7.0.6/kernel/sound/pci/ac97/snd-ac97-codec.ko
INSTALL /lib/modules/7.0.6/kernel/sound/ac97_bus.ko
INSTALL /lib/modules/7.0.6/kernel/net/802/psnap.ko
INSTALL /lib/modules/7.0.6/kernel/net/802/stp.ko
INSTALL /lib/modules/7.0.6/kernel/net/802/garp.ko
INSTALL /lib/modules/7.0.6/kernel/net/802/mrp.ko
INSTALL /lib/modules/7.0.6/kernel/net/ipv6/ipv6.ko
INSTALL /lib/modules/7.0.6/kernel/net/8021q/8021q.ko
INSTALL /lib/modules/7.0.6/kernel/net/wireless/cfg80211.ko
INSTALL /lib/modules/7.0.6/kernel/net/l1c/l1c.ko
INSTALL /lib/modules/7.0.6/kernel/net/rfkill/rfkill.ko
DEPMOD /lib/modules/7.0.6
root@slack:/usr/src/linux-7.0.6#

```

Potem należy zainstalować pliki nagłówkowe, za pomocą komendy `make headers_install`.

```

root@slack:/usr/src/linux-7.0.6# make headers_install
HOSTCC scripts/unifdef
HDRINST usr/include/asm-generic/fcntl.h
HDRINST usr/include/asm-generic/mman-common.h
HDRINST usr/include/asm-generic/auxvec.h
HDRINST usr/include/asm-generic/int-l164.h
HDRINST usr/include/asm-generic/int-l64.h
HDRINST usr/include/asm-generic/ipcbuf.h
HDRINST usr/include/asm-generic/types.h
HDRINST usr/include/asm-generic/signal.h
HDRINST usr/include/asm-generic/termbits-common.h
HDRINST usr/include/asm-generic/ioctls.h
HDRINST usr/include/asm-generic/bpf_perf_event.h
HDRINST usr/include/asm-generic/stat.h
HDRINST usr/include/asm-generic/posix_types.h
HDRINST usr/include/asm-generic/msgbuf.h
HDRINST usr/include/asm-generic/errno-base.h
HDRINST usr/include/asm-generic/termbits.h
HDRINST usr/include/asm-generic/mman.h
HDRINST usr/include/asm-generic/hugetlb_encode.h

```

```
HDRINST usr/include/linux/netfilter_bridge/ebt_vlan.h
HDRINST usr/include/linux/netfilter_bridge/ebt_pkttype.h
HDRINST usr/include/linux/netfilter_bridge/ebt_nat.h
HDRINST usr/include/linux/netfilter_bridge/ebt_mark_t.h
HDRINST usr/include/linux/netfilter_bridge/ebt_ip.h
HDRINST usr/include/linux/netfilter_bridge/ebt_redirect.h
HDRINST usr/include/linux/netfilter_bridge/ebt_among.h
HDRINST usr/include/linux/auxvec.h
HDRINST usr/include/linux/mps_iptunnel.h
HDRINST usr/include/linux/a.out.h
HDRINST usr/include/linux/serio.h
HDRINST usr/include/linux/matroxfb.h
HDRINST usr/include/linux/hdreg.h
HDRINST usr/include/linux/vm_sockets.h
HDRINST usr/include/linux/userfaultfd.h
HDRINST usr/include/linux/gtp.h
HDRINST usr/include/linux/nilfs2_ondisk.h
HDRINST usr/include/linux/seg6_local.h
HDRINST usr/include/linux/v4l2-mediabus.h
HDRINST usr/include/linux/rds.h
HDRINST usr/include/linux/udp.h
HDRINST usr/include/linux/atm.h
HDRINST usr/include/linux/gen_stats.h
HDRINST usr/include/linux/atm_eni.h
HDRINST usr/include/linux/arcfb.h
HDRINST usr/include/linux/bpf_common.h
HDRINST usr/include/linux/netfilter_arp/arp_tables.h
HDRINST usr/include/linux/netfilter_arp/arpt_mangle.h
HDRINST usr/include/linux/smc.h
HDRINST usr/include/linux/sched/types.h
HDRINST usr/include/linux/cryptouser.h
HDRINST usr/include/linux/rpl_iptunnel.h
HDRINST usr/include/linux/tipc_sockets_diag.h
HDRINST usr/include/linux/auto_fs.h
HDRINST usr/include/linux/fsi.h
HDRINST usr/include/linux/utime.h
HDRINST usr/include/linux/igmp.h
HDRINST usr/include/linux/atmarp.h
HDRINST usr/include/linux/joystick.h
HDRINST usr/include/linux/fsl_mc.h
HDRINST usr/include/linux/papr_pdsn.h
HDRINST usr/include/linux/kd.h
HDRINST usr/include/linux/virtio_scsi.h
HDRINST usr/include/linux/major.h
HDRINST usr/include/linux/times.h
HDRINST usr/include/linux/netfilter.h
HDRINST usr/include/linux/hw_breakpoint.h
```

```

HDRINST usr/include/asm/mman.h
HDRINST usr/include/asm/msr.h
HDRINST usr/include/asm/setup_data.h
HDRINST usr/include/asm/sembuf.h
HDRINST usr/include/asm/ucontext.h
HDRINST usr/include/asm/processor-flags.h
HDRINST usr/include/asm/swab.h
HDRINST usr/include/asm/amd_hsmpl.h
HDRINST usr/include/asm/statfs.h
HDRINST usr/include/asm/ptrace-abi.h
HDRINST usr/include/asm/setup.h
HDRINST usr/include/asm/bootparam.h
HDRINST usr/include/asm/sgx.h
HDRINST usr/include/asm/prctl.h
HDRINST usr/include/asm/kvm_para.h
HDRINST usr/include/asm/ist.h
HDRINST usr/include/asm/bitsperlong.h
HDRINST usr/include/asm/boot.h
HDRINST usr/include/asm/mce.h
HDRINST usr/include/asm/siginfo.h
HDRINST usr/include/asm/svm.h
HDRINST usr/include/asm/shmbuf.h
HDRINST usr/include/asm/perf_regs.h
HDRINST usr/include/asm/posix_types_64.h
HDRINST usr/include/asm/vm86.h
HDRINST usr/include/asm/unistd.h
HDRINST usr/include/asm/usyscall.h
HDRINST usr/include/asm/vmx.h
HDRINST usr/include/asm/sigcontext32.h
HDRINST usr/include/asm/fcntl.h
HDRINST usr/include/asm/unistd_x32.h
HDRINST usr/include/asm/ipcbuf.h
HDRINST usr/include/asm/types.h
HDRINST usr/include/asm/ioctls.h
HDRINST usr/include/asm/bpf_perf_event.h
HDRINST usr/include/asm/termbits.h
HDRINST usr/include/asm/socket.h
HDRINST usr/include/asm/resource.h
HDRINST usr/include/asm/poll.h
HDRINST usr/include/asm/unistd_64.h
HDRINST usr/include/asm/ioctl.h
HDRINST usr/include/asm/termios.h
HDRINST usr/include/asm/unistd_32.h
HDRINST usr/include/asm/sockios.h
HDRINST usr/include/asm/errno.h
HDRINST usr/include/asm/param.h
INSTALL ./usr/include
root@slack:/usr/src/linux-7.0.6#

```

Kolejny krok to sprawdzenie zainstalowanych modułów w katalogu /lib/modules/7.0.6.

```

root@slack:/usr/src/linux-7.0.6# ls /lib/modules/7.0.6/
build@      modules.alias.bin      modules.builtin.bin    modules.dep.bin        modules.softdep
kernel/     modules.builtin        modules.builtin.modinfo modules.devname         modules.symbols
modules.alias modules.builtin.alias.bin modules.dep             modules.order          modules.symbols.bin

```

Następnie należy skopiować pliki kernela do systemu do katalogu /boot, za pomocą polecenia: `cp arch/x86_64/boot/bzImage /boot/vmlinuz-custom-7.0.6`.

```

root@slack:/usr/src/linux-7.0.6# cp arch/x86_64/boot/bzImage /boot/vmlinuz-custom-7.0.6

```

Później trzeba przekopiować tablicę symboli używanych przez kernel, za pomocą komendy: `cp System.map /boot/System.map-custom-7.0.6`.

```

root@slack:/usr/src/linux-7.0.6# cp System.map /boot/System.map-custom-7.0.6

```


Potem należy przekopiować plik konfiguracyjny kernela, za pomocą polecenia:
`cp .config /boot/config-custom-7.0.6.`

```
root@slack:/usr/src/linux-7.0.6# cp .config /boot/config-custom-7.0.6
```

Następnie należy przejść do katalogu /boot, usunąć starą tablicę symboli i utworzyć link symboliczny dla tablicy symboli kernela.

```
root@slack:/usr/src/linux-7.0.6# cd /boot
root@slack:/boot# rm System.map
root@slack:/boot# ln -s System.map-custom-7.0.6 System.map
```

Kolejny krok to utworzenie dysku RAM, za pomocą wywołania skryptu i wykorzystania komendy, wygenerowanej przez ten skrypt.

```
root@slack:/boot# /usr/share/mkinitrd/mkinitrd_command_generator.sh -k 7.0.6
#
# mkinitrd_command_generator.sh revision 1.45
#
# This script will now make a recommendation about the command to use
# in case you require an initrd image to boot a kernel that does not
# have support for your storage or root filesystem built in
# (such as the Slackware 'generic' kernels').
# A suitable 'mkinitrd' command will be:

mkinitrd -c -k 7.0.6 -f ext4 -r /dev/sda1 -m jbd2:mbcache:crc16:ext4 -u -o /boot/initrd.gz
root@slack:/boot# mkinitrd -c -k 7.0.6 -f ext4 -r /dev/sda1 -m jbd2:mbcache:crc16:ext4 -u -o /boot/initrd-custom-7.0.6.gz
OK: /lib/modules/7.0.6/kernel/fs/jbd2/jbd2.ko added.
OK: /lib/modules/7.0.6/kernel/fs/mbcache.ko added.
OK: /lib/modules/7.0.6/kernel/lib/crc/crc16.ko added.
OK: /lib/modules/7.0.6/kernel/fs/jbd2/jbd2.ko added.
OK: /lib/modules/7.0.6/kernel/fs/mbcache.ko added.
OK: /lib/modules/7.0.6/kernel/lib/crc/crc16.ko added.
OK: /lib/modules/7.0.6/kernel/fs/ext4/ext4.ko added.
55043 blocks
/boot/initrd-custom-7.0.6.gz created.
Be sure to run lilo again if you use it.
```

Później trzeba ponownie skonfigurować GRUB.

```
root@slack:/boot# grub-mkconfig -o /boot/grub/grub.cfg
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-huge-5.15.19
Found initrd image: /boot/initrd.gz
Found linux image: /boot/vmlinuz-huge
Found initrd image: /boot/initrd.gz
Found linux image: /boot/vmlinuz-generic-5.15.19
Found initrd image: /boot/initrd.gz
Found linux image: /boot/vmlinuz-generic
Found initrd image: /boot/initrd.gz
Found linux image: /boot/vmlinuz-custom-7.0.6
Found initrd image: /boot/initrd.gz
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for EFI Firmware Settings ...
done
root@slack:/boot#
```

Następnie należy skopiować obraz jądra oraz ramdisku na partycję EFI.

```
root@slack:/boot# cp /boot/vmlinuz-custom-7.0.6 /boot/efi/EFI/Slackware/
root@slack:/boot# cp /boot/initrd-custom-7.0.6.gz /boot/efi/EFI/Slackware/
```

Kolejnym krokiem jest dodanie nowego wpisu do konfiguracji bootloadera ELILO.

```
root@slack:/boot# nano /boot/efi/EFI/Slackware/elilo.conf
```

```

GNU nano 6.0 /boot/efi/EFI/Slackware/elilo.conf
#chooser=simple
prompt
delay=15
timeout=30
default=custom-kernel-7.0.6
#
image=vmlinuz
    label=vmlinuz
    initrd=initrd.gz
    read-only
    append="root=/dev/sda1 uga=normal ro"

image=vmlinuz-custom-7.0.6
    label=custom-kernel-7.0.6
    initrd=initrd-custom-7.0.6.gz
    read-only
    append="root=/dev/sda1 uga=791 ro"

```

Następnie należy zrestartować system, za pomocą polecenia reboot i w opcjach zaawansowanych wybrać odpowiedni kernel.

```

root@slack:/boot# reboot

Broadcast message from root@slack.localhost (tty1) (Thu May 14 10:00:16 2026):

The system is going down for reboot NOW!
INIT: Switching to runlevel: 6
INIT: Sending processes configured via /etc/inittab the TERM signal

Running shutdown script /etc/rc.d/rc.6:
INIT: version 3.01 reloading
Saving system time to the hardware clock (localtime).
Stopping haveged.
Unmounting remote filesystems:
Stopping system message bus...
Stopping the network interfaces...
Stopping udevd
Sending all processes the SIGTERM signal.
[ 4105.309097] elogind-daemon[1035]: Received signal 15 [TERM]

```

GNU GRUB version 2.06

Slackware-15.0 GNU/Linux
*Advanced options for Slackware-15.0 GNU/Linux
UEFI Firmware Settings

Use the ▲ and ▼ keys to select which entry is highlighted.
Press enter to boot the selected OS, 'e' to edit the commands before booting or 'c' for a command-line.

GNU GRUB version 2.06

Slackware-15.0 GNU/Linux, with Linux 5.15.19
Slackware-15.0 GNU/Linux, with Linux 5.15.19 (recovery mode)
Slackware-15.0 GNU/Linux, with Linux huge
Slackware-15.0 GNU/Linux, with Linux huge (recovery mode)
Slackware-15.0 GNU/Linux, with Linux 5.15.19
Slackware-15.0 GNU/Linux, with Linux 5.15.19 (recovery mode)
Slackware-15.0 GNU/Linux, with Linux generic
Slackware-15.0 GNU/Linux, with Linux generic (recovery mode)
*Slackware-15.0 GNU/Linux, with Linux 7.0.6
Slackware-15.0 GNU/Linux, with Linux 7.0.6 (recovery mode)

Use the ▲ and ▼ keys to select which entry is highlighted.
Press enter to boot the selected OS, 'e' to edit the commands before booting or 'c' for a command-line. ESC to return previous menu.

Kernel spanikował, tak jak na poniższym zdjęciu.



```
/bin/sh: can't access tty: job control turned off
/ # exit
mount: mounting /proc on /mnt/proc failed: No such file or directory
mount: mounting /sys on /mnt/sys failed: No such file or directory
mount: mounting /run on /mnt/run failed: No such file or directory
mount: mounting /dev on /mnt/dev failed: No such file or directory
/boot/initrd.gz: exiting
BusyBox v1.32.1 (2022-01-26 14:33:14 CST) multi-call binary.

Usage: switch_root [-c CONSOLE_DEV] NEW_ROOT NEW_INIT [ARGS]

Free initramfs and switch to another root fs:
chroot to NEW_ROOT, delete all in /, move NEW_ROOT to /,
execute NEW_INIT. PID must be 1. NEW_ROOT must be a mountpoint.

-c DEV Reopen stdio to DEV after switch
[ 42.059691] Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000100
[ 42.059788] CPU: 0 UID: 0 PID: 1 Comm: switch_root Not tainted 7.0.6 #1 PREEMPT(full)
[ 42.059883] Hardware name: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
[ 42.059975] Call Trace:
[ 42.060066] <TASK>
[ 42.060157] upanic+0x455/0x4b0
[ 42.060248] panic+0x5b/0x60
[ 42.060332] do_exit+0x8fe/0xa80
[ 42.060416] ? count_memcg_events+0x77/0x1a0
[ 42.060503] do_group_exit+0x30/0x80
[ 42.060584] __x64_sys_exit_group+0x18/0x20
[ 42.060666] x64_sys_call+0x18d3/0x18e0
[ 42.060768] do_syscall_64+0x9e/0x490
[ 42.060855] entry_SYSCALL_64_after_hwframe+0x76/0x7e
[ 42.060936] RIP: 0033:0x7f290d242879
[ 42.061017] Code: 00 4c 8b 05 f1 05 10 00 be e7 00 00 00 ba 3c 00 00 00 eb 12 0f 1f 44 00 00 89 d0 0f 05 48 3d 00 f0 ff ff 7
1c f4 89 f0 0f 05 <48> 3d 00 f0 ff ff 76 e7 f7 d8 64 41 89 00 eb df 0f 1f 80 00 00 00
[ 42.061190] RSP: 002b:00007ffec13bdd38 EFLAGS: 00000246 ORIG_RAX: 00000000000000e7
[ 42.061278] RAX: ffffffff8bda RBX: 00007f290d33f920 RCX: 00007f290d242879
[ 42.061366] RDX: 000000000000003c RSI: 00000000000000e7 RDI: 0000000000000001
[ 42.061455] RBP: 0000000000000001 R08: ffffffff8b R09: 0000000000000001
[ 42.061544] R10: 00007f290d17ee70 R11: 0000000000000246 R12: 00007f290d33f920
[ 42.061630] R13: 0000000000000001 R14: 00007f290d344e48 R15: 0000000000000000
[ 42.061717] </TASK>
[ 42.062322] Kernel Offset: 0x1f000000 from 0xffffffff81000000 (relocation range: 0xffffffff80000000-0xffffffffbfffffff)
[ 42.062417] ---[ end Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000100 ]---
```

W takim razie należy zmienić 2 wpisy w pliku /boot/grub/grub.cfg ze /boot/initrd.gz na /boot/initrd-custom-7.0.6.gz.

```

GNU nano 6.0 /boot/grub/grub.cfg Modified
if [ x$feature_platform_search_hint = xy ]; then
  search --no-floppy --fs-uuid --set=root --hint-bios=hd0,gpt1 --hint-efi=hd0,gpt1 --hint-baremetal=ahci0,gpt1
else
  search --no-floppy --fs-uuid --set=root a4e7d1cf-fe44-4734-aaea-d136e21ec10a
fi
echo 'Loading Linux generic ...'
linux /boot/vmlinuz-generic root=UUID=a4e7d1cf-fe44-4734-aaea-d136e21ec10a ro single
echo 'Loading initial ramdisk ...'
initrd /boot/initrd.gz
}
menuentry 'Slackware-15.0 GNU/Linux, with Linux 7.0.6' --class slackware_15_0 --class gnu-linux --class gnu --class os
{
  load_video
  insmod gzio
  insmod part_gpt
  insmod ext2
  set root='hd0,gpt1'
  if [ x$feature_platform_search_hint = xy ]; then
    search --no-floppy --fs-uuid --set=root --hint-bios=hd0,gpt1 --hint-efi=hd0,gpt1 --hint-baremetal=ahci0,gpt1
  else
    search --no-floppy --fs-uuid --set=root a4e7d1cf-fe44-4734-aaea-d136e21ec10a
  fi
  echo 'Loading Linux 7.0.6 ...'
  linux /boot/vmlinuz-custom-7.0.6 root=UUID=a4e7d1cf-fe44-4734-aaea-d136e21ec10a ro
  echo 'Loading initial ramdisk ...'
  initrd /boot/initrd-custom-7.0.6.gz
}
menuentry 'Slackware-15.0 GNU/Linux, with Linux 7.0.6 (recovery mode)' --class slackware_15_0 --class gnu-linux --class gnu --class os
{
  load_video
  insmod gzio
  insmod part_gpt
  insmod ext2
  set root='hd0,gpt1'
  if [ x$feature_platform_search_hint = xy ]; then
    search --no-floppy --fs-uuid --set=root --hint-bios=hd0,gpt1 --hint-efi=hd0,gpt1 --hint-baremetal=ahci0,gpt1
  else
    search --no-floppy --fs-uuid --set=root a4e7d1cf-fe44-4734-aaea-d136e21ec10a
  fi
  echo 'Loading Linux 7.0.6 ...'
  linux /boot/vmlinuz-custom-7.0.6 root=UUID=a4e7d1cf-fe44-4734-aaea-d136e21ec10a ro single
  echo 'Loading initial ramdisk ...'
  initrd /boot/initrd-custom-7.0.6.gz
}
}

```

Następnie należy zrestartować system, za pomocą polecenia reboot, wybrać odpowiedni kernel jak wcześniej i czekać z nadzieją, że zadziała.

JEST SUKCES!!! Uff... po zmianach w pliku /boot/grub/grub.cfg ładuje się odpowiedni kernel.

```

/dev/sda4 on /boot/efi type efat (rw,relatime,fmask=0022,dmask=0022,codepage=437,ioccharset=iso8859-1,shortname=mixed,utf8,errors=remount-ro)
Using /etc/random-seed to initialize /dev/urandom.
INIT: Entering runlevel: 3
Going multiuser...
Updating shared library links: /sbin/ldconfig &
Starting syslogd daemon: /usr/sbin/syslogd -s
Updating X font indexes: /usr/bin/fc-cache -f &
Updating hardware database index: /sbin/udevadm hwdb --update
Triggering udev events: /sbin/udevadm trigger --action=change
Starting haveged entropy daemon: /sbin/haveged
Starting the network interfaces...
eth0: polling for DHCP server
dhcpcd-9.4.1 starting
DUID 00:04:3b:3a:a8:d0:3b:2b:bb:4e:b2:67:99:51:2a:21:85:a4
eth0: waiting for carrier
eth0: carrier acquired
eth0: IAD 27:88:e6:96
eth0: soliciting a DHCP lease
eth0: offered 10.0.2.15 from 10.0.2.2
eth0: probing address 10.0.2.15/24
eth0: leased 10.0.2.15 for 86400 seconds
eth0: adding route to 10.0.2.0/24
eth0: adding default route via 10.0.2.2
forked to background, child pid 510
Starting system message bus: /usr/bin/dbus-uuidgen --ensure ; /usr/bin/dbus-daemon --system
Starting elogind: /lib64/elogind/elogind --daemon
Starting OpenSSH SSH daemon: /usr/sbin/sshd
Starting ACPI daemon: /usr/sbin/acpid
Updating MIME database: /usr/bin/update-mime-database /usr/share/mime &
Updating gtk.immodules:
  /usr/bin/update-gtk-immodules &
Updating gdk-pixbuf.loaders:
  /usr/bin/update-gdk-pixbuf-loaders &
Compiling GSettings XML schema files:
  /usr/bin/glib-compile-schemas /usr/share/glib-2.0/schemas &
Starting crond: /usr/sbin/crond -l notice
Starting atd: /usr/sbin/atd -b 15 -l 1
Loading /usr/share/kbd/keymaps/i386/qlerty/pl.map.gz
Starting gpm: /usr/sbin/gpm -m /dev/mouse -t imps2

Welcome to Linux 7.0.6 x86_64 (tty1)

slack login: root
Password:
Last login: Thu May 14 10:19:14 on tty1
Linux 7.0.6.
root@slack:~# uname -r
7.0.6
root@slack:~#

```

Kompilacja jądra z wykorzystaniem streamline_config.pl

Na początku należy przejść do katalogu /usr/src/, za pomocą komendy cd i pobrać najnowsze źródła kernela, wykorzystując wget link_do_pobrania_kernelu.

```

root@slack:~# cd /usr/src
root@slack:/usr/src# wget https://cdn.kernel.org/pub/linux/kernel/v7.x/linux-7.0.6.tar.xz
--2026-05-14 13:02:47-- https://cdn.kernel.org/pub/linux/kernel/v7.x/linux-7.0.6.tar.xz
Resolving cdn.kernel.org (cdn.kernel.org)... 151.101.5.176, 2a04:4e42:1::432
Connecting to cdn.kernel.org (cdn.kernel.org)|151.101.5.176|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 157158988 (150M) [application/x-xz]
Saving to: 'linux-7.0.6.tar.xz'

linux-7.0.6.tar.xz          100%[=====>] 149.88M  43.3MB/s   in 4.1s

2026-05-14 13:02:52 (36.1 MB/s) - 'linux-7.0.6.tar.xz' saved [157158988/157158988]

root@slack:/usr/src#

```

Następnie należy rozpakować plik archiwum ze źródłami kernela, za pomocą komendy tar z opcjami xvpf.

```

root@slack:/usr/src# tar -xvpf linux-7.0.6.tar.xz

```

```

linux-7.0.6/tools/virtio/xen/xen.h
linux-7.0.6/tools/wmi/
linux-7.0.6/tools/wmi/Makefile
linux-7.0.6/tools/wmi/dell-smbios-example.c
linux-7.0.6/tools/workqueue/
linux-7.0.6/tools/workqueue/wq_dump.py
linux-7.0.6/tools/workqueue/wq_monitor.py
linux-7.0.6/tools/writeback/
linux-7.0.6/tools/writeback/wb_monitor.py
linux-7.0.6/usr/
linux-7.0.6/usr/.gitignore
linux-7.0.6/usr/Kconfig
linux-7.0.6/usr/Makefile
linux-7.0.6/usr/default_cpio_list
linux-7.0.6/usr/dummy-include/
linux-7.0.6/usr/dummy-include/stdbool.h
linux-7.0.6/usr/dummy-include/stdlib.h
linux-7.0.6/usr/gen_init_cpio.c
linux-7.0.6/usr/gen_initramfs.sh
linux-7.0.6/usr/include/
linux-7.0.6/usr/include/.gitignore
linux-7.0.6/usr/include/Makefile
linux-7.0.6/usr/include/headers_check.pl
linux-7.0.6/usr/initramfs_data.S
linux-7.0.6/virt/
linux-7.0.6/virt/Makefile
linux-7.0.6/virt/kvm/
linux-7.0.6/virt/kvm/Kconfig
linux-7.0.6/virt/kvm/Makefile.kvm
linux-7.0.6/virt/kvm/async_pf.c
linux-7.0.6/virt/kvm/async_pf.h
linux-7.0.6/virt/kvm/binary_stats.c
linux-7.0.6/virt/kvm/coalesced_mmio.c
linux-7.0.6/virt/kvm/coalesced_mmio.h
linux-7.0.6/virt/kvm/dirty_ring.c
linux-7.0.6/virt/kvm/eventfd.c
linux-7.0.6/virt/kvm/guest_memfd.c
linux-7.0.6/virt/kvm/irqchip.c
linux-7.0.6/virt/kvm/kvm_main.c
linux-7.0.6/virt/kvm/kvm_mm.h
linux-7.0.6/virt/kvm/pfncache.c
linux-7.0.6/virt/kvm/ufio.c
linux-7.0.6/virt/kvm/ufio.h
linux-7.0.6/virt/lib/
linux-7.0.6/virt/lib/Kconfig
linux-7.0.6/virt/lib/Makefile
linux-7.0.6/virt/lib/irqbypass.c
root@slack:/usr/src#

```

Potem trzeba przejść do folderu z rozpakowanym kernelem.

```

linux-7.0.6/virt/kvm/ufio.h
linux-7.0.6/virt/lib/
linux-7.0.6/virt/lib/Kconfig
linux-7.0.6/virt/lib/Makefile
linux-7.0.6/virt/lib/irqbypass.c
root@slack:/usr/src# ls
linux@ linux-5.15.19/ linux-7.0.6/ linux-7.0.6.tar.xz
root@slack:/usr/src# cd linux-7.0.6
root@slack:/usr/src/linux-7.0.6#

```

Kolejny krok to skopiowanie konfiguracji starego jądra, za pomocą polecenia `zcat`.

```

root@slack:/usr/src/linux-7.0.6# zcat /proc/config.gz > .config
root@slack:/usr/src/linux-7.0.6# ls -al
total 1548
drwxrwxr-x 26 root root 4096 May 14 13:12 ./
drwxr-xr-x 4 root root 4096 May 14 13:07 ../
-rw-rw-r-- 1 root root 24291 May 11 08:21 .clang-format
-rw-rw-r-- 1 root root 374 May 11 08:21 .clippy.toml
-rw-rw-r-- 1 root root 59 May 11 08:21 .cocciconfig
-rw-rw-r-- 1 root root 239253 May 14 13:12 .config
-rw-rw-r-- 1 root root 566 May 11 08:21 .editorconfig
-rw-rw-r-- 1 root root 305 May 11 08:21 .get_maintainer.ignore
-rw-rw-r-- 1 root root 105 May 11 08:21 .gitattributes
-rw-rw-r-- 1 root root 2238 May 11 08:21 .gitignore
-rw-rw-r-- 1 root root 54512 May 11 08:21 .mailmap
-rw-rw-r-- 1 root root 66 May 11 08:21 .pylintrc
-rw-rw-r-- 1 root root 369 May 11 08:21 .rustfmt.toml
-rw-rw-r-- 1 root root 496 May 11 08:21 COPYING
-rw-rw-r-- 1 root root 107789 May 11 08:21 CREDITS
drwxrwxr-x 78 root root 4096 May 11 08:21 Documentation/
-rw-rw-r-- 1 root root 2871 May 11 08:21 Kbuild
-rw-rw-r-- 1 root root 582 May 11 08:21 Kconfig
drwxrwxr-x 6 root root 4096 May 11 08:21 LICENSES/
-rw-rw-r-- 1 root root 894639 May 11 08:21 MAINTAINERS
-rw-rw-r-- 1 root root 74251 May 11 08:21 Makefile
-rw-rw-r-- 1 root root 6043 May 11 08:21 README
drwxrwxr-x 23 root root 4096 May 11 08:21 arch/
drwxrwxr-x 3 root root 4096 May 11 08:21 block/
drwxrwxr-x 2 root root 4096 May 11 08:21 certs/
drwxrwxr-x 5 root root 4096 May 11 08:21 crypto/
drwxrwxr-x 147 root root 4096 May 11 08:21 drivers/
drwxrwxr-x 79 root root 4096 May 11 08:21 fs/
drwxrwxr-x 33 root root 4096 May 11 08:21 include/
drwxrwxr-x 2 root root 4096 May 11 08:21 init/
drwxrwxr-x 2 root root 4096 May 11 08:21 io_uring/
drwxrwxr-x 2 root root 4096 May 11 08:21 ipc/
drwxrwxr-x 24 root root 4096 May 11 08:21 kernel/
drwxrwxr-x 22 root root 12288 May 11 08:21 lib/
drwxrwxr-x 7 root root 4096 May 11 08:21 mm/
drwxrwxr-x 73 root root 4096 May 11 08:21 net/
drwxrwxr-x 11 root root 4096 May 11 08:21 rust/
drwxrwxr-x 47 root root 4096 May 11 08:21 samples/
drwxrwxr-x 23 root root 12288 May 11 08:21 scripts/
drwxrwxr-x 15 root root 4096 May 11 08:21 security/
drwxrwxr-x 27 root root 4096 May 11 08:21 sound/
drwxrwxr-x 47 root root 4096 May 11 08:21 tools/
drwxrwxr-x 4 root root 4096 May 11 08:21 usr/
drwxrwxr-x 4 root root 4096 May 11 08:21 virt/
root@slack:/usr/src/linux-7.0.6#

```

Następnie należy sprawdzić skrypt do tworzenia pliku .config tylko z tymi modułami , które używamy.

```

root@slack:/usr/src/linux-7.0.6# nano scripts/kconfig/streamline_config.pl

```

Potem należy uruchomić skrypt. Tu: wynik skryptu jest zapisywana do pliku config_strip.

```

root@slack:/usr/src/linux-7.0.6# scripts/kconfig/streamline_config.pl > config_strip
using config: '.config'
crt10dif_pclmul config not found!
crc32_pclmul config not found!
module fb_sys_fops did not have configs CONFIG_FB_SYSMEM_FOPS
root@slack:/usr/src/linux-7.0.6#

```

Kolejnym krokiem jest zamiana wcześniej utworzonego pliku config_strip na plik konfiguracyjny.


```

root@slack:/usr/src/linux-7.0.6# make KCONFIG_ALLCONFIG=config_strip olddefconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/kconfig/conf.o
HOSTCC scripts/kconfig/confdata.o
HOSTCC scripts/kconfig/expr.o
LEX scripts/kconfig/lexer.lex.c
YACC scripts/kconfig/parser.tab.[ch]
HOSTCC scripts/kconfig/lexer.lex.o
HOSTCC scripts/kconfig/menu.o
HOSTCC scripts/kconfig/parser.tab.o
HOSTCC scripts/kconfig/preprocess.o
HOSTCC scripts/kconfig/symbol.o
HOSTCC scripts/kconfig/util.o
HOSTLD scripts/kconfig/conf
.config:429:warning: symbol value 'm' invalid for IBK
.config:838:warning: symbol value '0' invalid for BASE_SMALL
.config:3072:warning: symbol value 'm' invalid for NET_DSA_REALTEK_SMI
.config:6287:warning: symbol value 'm' invalid for FB_BACKLIGHT
.config:7237:warning: symbol value 'm' invalid for USB_FOTG210_HCD
.config:7880:warning: symbol value 'm' invalid for VFIO_VIRQFD
.config:7930:warning: symbol value 'm' invalid for HYPERV
.config:8028:warning: symbol value 'm' invalid for VIDEO_ZORAN_DC30
.config:8029:warning: symbol value 'm' invalid for VIDEO_ZORAN_ZR36060
.config:8030:warning: symbol value 'm' invalid for VIDEO_ZORAN_BUZ
.config:8031:warning: symbol value 'm' invalid for VIDEO_ZORAN_DC10
.config:8032:warning: symbol value 'm' invalid for VIDEO_ZORAN_LML33
.config:8033:warning: symbol value 'm' invalid for VIDEO_ZORAN_LML33R10
.config:8034:warning: symbol value 'm' invalid for VIDEO_ZORAN_AVS6EYES
.config:9151:warning: symbol value 'm' invalid for FSCACHE
.config:9692:warning: symbol value 'm' invalid for CRYPTO_LIB_CURVE25519_GENERIC
.config:9697:warning: symbol value 'm' invalid for CRYPTO_LIB_POLY1305_GENERIC
#
# configuration written to .config
#
root@slack:/usr/src/linux-7.0.6#

```

Później należy zbudować plik konfiguracyjny kernela z konfiguracją z poprzedniego kernela, za pomocą polecenia `make olddefconfig`.

```

root@slack:/usr/src/linux-7.0.6# make olddefconfig
#
# No change to .config
#
root@slack:/usr/src/linux-7.0.6#

```

Kolejnym krokiem jest skompilowanie obrazu jądra, za pomocą polecenia `make bzImage`.

```

root@slack:/usr/src/linux-7.0.6# make bzImage
SYSHDR arch/x86/include/generated/uapi/asm/unistd_32.h
SYSHDR arch/x86/include/generated/uapi/asm/unistd_64.h
SYSHDR arch/x86/include/generated/uapi/asm/unistd_x32.h
SYSTBL arch/x86/include/generated/asm/syscalls_32.h
SYSHDR arch/x86/include/generated/asm/unistd_32_ia32.h
SYSHDR arch/x86/include/generated/asm/unistd_64_x32.h
SYSTBL arch/x86/include/generated/asm/syscalls_64.h
HOSTCC arch/x86/tools/relocs_32.o
HOSTCC arch/x86/tools/relocs_64.o
HOSTCC arch/x86/tools/relocs_common.o
HOSTLD arch/x86/tools/relocs
HOSTCC arch/x86/tools/vdso2c

```

```
HOSTLD scripts/sorttable
HOSTCC scripts/asn1_compiler
HOSTCC scripts/tracepoint-update.o
HOSTLD scripts/tracepoint-update
WRAP arch/x86/include/generated/uapi/asm/bpf_perf_event.h
WRAP arch/x86/include/generated/uapi/asm/errno.h
WRAP arch/x86/include/generated/uapi/asm/fcntl.h
WRAP arch/x86/include/generated/uapi/asm/ioctl.h
WRAP arch/x86/include/generated/uapi/asm/ioctls.h
WRAP arch/x86/include/generated/uapi/asm/ipcbuf.h
WRAP arch/x86/include/generated/uapi/asm/param.h
WRAP arch/x86/include/generated/uapi/asm/poll.h
WRAP arch/x86/include/generated/uapi/asm/resource.h
WRAP arch/x86/include/generated/uapi/asm/socket.h
WRAP arch/x86/include/generated/uapi/asm/sockios.h
WRAP arch/x86/include/generated/uapi/asm/termbits.h
WRAP arch/x86/include/generated/uapi/asm/termios.h
WRAP arch/x86/include/generated/uapi/asm/types.h
WRAP arch/x86/include/generated/asm/early_ioremap.h
WRAP arch/x86/include/generated/asm/fprobe.h
WRAP arch/x86/include/generated/asm/mcs_spinlock.h
WRAP arch/x86/include/generated/asm/mmzone.h
WRAP arch/x86/include/generated/asm/irq_regs.h
WRAP arch/x86/include/generated/asm/kmap_size.h
WRAP arch/x86/include/generated/asm/local64.h
WRAP arch/x86/include/generated/asm/mmiowb.h
WRAP arch/x86/include/generated/asm/module.lds.h
WRAP arch/x86/include/generated/asm/rwonce.h
UPD arch/x86/include/generated/asm/cpufeaturemasks.h
GEN arch/x86/include/generated/asm/orc_hash.h
UPD include/config/kernel.release
UPD include/generated/uapi/linux/version.h
UPD include/generated/utsrelease.h
UPD include/generated/compile.h
CC scripts/mod/empty.o
HOSTCC scripts/mod/mk_elfconfig
MKELF scripts/mod/elfconfig.h
HOSTCC scripts/mod/modpost.o
CC scripts/mod/devicetable-offsets.s
UPD scripts/mod/devicetable-offsets.h
HOSTCC scripts/mod/file2alias.o
HOSTCC scripts/mod/sumversion.o
HOSTCC scripts/mod/symsearch.o
HOSTLD scripts/mod/modpost
UPD include/generated/timeconst.h
CC kernel/bounds.s
UPD include/generated/bounds.h
```

```

CC      arch/x86/kernel/cpu/scattered.o
CC      arch/x86/kernel/cpu/topology_common.o
CC      arch/x86/kernel/cpu/topology_ext.o
CC      arch/x86/kernel/cpu/topology_amd.o
CC      arch/x86/kernel/cpu/common.o
CC      arch/x86/kernel/cpu/rdrand.o
CC      arch/x86/kernel/cpu/match.o
CC      arch/x86/kernel/cpu/bugs.o
CC      arch/x86/kernel/cpu/aperfmpperf.o
CC      arch/x86/kernel/cpu/cpuid-deps.o
CC      arch/x86/kernel/cpu/cpuid_0x2_table.o
CC      arch/x86/kernel/cpu/unwait.o
MKCAP   arch/x86/kernel/cpu/capflags.c
CC      arch/x86/kernel/cpu/capflags.o
CC      arch/x86/kernel/cpu/powerflags.o
CC      arch/x86/kernel/cpu/topology.o
CC      arch/x86/kernel/cpu/proc.o
CC      arch/x86/kernel/cpu/feat_ctl.o
CC      arch/x86/kernel/cpu/intel.o
CC      arch/x86/kernel/cpu/tsx.o
CC      arch/x86/kernel/cpu/intel_epb.o
CC      arch/x86/kernel/cpu/amd.o
CC      arch/x86/kernel/cpu/amd_cache_disable.o
CC      arch/x86/kernel/cpu/hygon.o
CC      arch/x86/kernel/cpu/centaur.o
CC      arch/x86/kernel/cpu/zhaoxin.o
CC      arch/x86/kernel/cpu/perfctr-watchdog.o
CC      arch/x86/kernel/cpu/umware.o
CC      arch/x86/kernel/cpu/hypervisor.o
CC      arch/x86/kernel/cpu/mshyperu.o
CC      arch/x86/kernel/cpu/acrn.o
CC      arch/x86/kernel/cpu/debugfs.o
AR      arch/x86/kernel/cpu/built-in.a
CC      arch/x86/kernel/acpi/boot.o
CC      arch/x86/kernel/acpi/sleep.o
AS      arch/x86/kernel/acpi/wakeup_64.o
CC      arch/x86/kernel/acpi/apei.o
CC      arch/x86/kernel/acpi/cppc.o
CC      arch/x86/kernel/acpi/madt_wakeup.o
AS      arch/x86/kernel/acpi/madt_playdead.o
CC      arch/x86/kernel/acpi/cstate.o
AR      arch/x86/kernel/acpi/built-in.a
CC      arch/x86/kernel/apic/apic.o
CC      arch/x86/kernel/apic/apic_common.o
CC      arch/x86/kernel/apic/apic_noop.o
CC      arch/x86/kernel/apic/ipi.o
CC      arch/x86/kernel/apic/vector.o

```

```

AS      arch/x86/boot/compressed/head_64.o
VOFFSET arch/x86/boot/compressed/..voffset.h
CC      arch/x86/boot/compressed/misc.o
CC      arch/x86/boot/compressed/string.o
CC      arch/x86/boot/compressed/cmdline.o
CC      arch/x86/boot/compressed/error.o
OBJCOPY arch/x86/boot/compressed/vmlinux.bin
RELOCS  arch/x86/boot/compressed/vmlinux.relocs
LZMA    arch/x86/boot/compressed/vmlinux.bin.lzma
HOSTCC  arch/x86/boot/compressed/mkpiggy
MKPIGGY arch/x86/boot/compressed/piggy.S
AS      arch/x86/boot/compressed/piggy.o
CC      arch/x86/boot/compressed/cpuflags.o
CC      arch/x86/boot/compressed/early_serial_console.o
CC      arch/x86/boot/compressed/kaslr.o
CC      arch/x86/boot/compressed/ident_map_64.o
CC      arch/x86/boot/compressed/idt_64.o
AS      arch/x86/boot/compressed/idt_handlers_64.o
AS      arch/x86/boot/compressed/mem_encrypt.o
CC      arch/x86/boot/compressed/pgtable_64.o
CC      arch/x86/boot/compressed/sev.o
CC      arch/x86/boot/compressed/sev-handle-vc.o
CC      arch/x86/boot/compressed/acpi.o
CC      arch/x86/boot/compressed/mem.o
CC      arch/x86/boot/compressed/efi.o
LD      arch/x86/boot/compressed/vmlinux
ZOFFSET arch/x86/boot/zoffset.h
AS      arch/x86/boot/header.o
CC      arch/x86/boot/main.o
CC      arch/x86/boot/memory.o
CC      arch/x86/boot/pm.o
AS      arch/x86/boot/pmjump.o
CC      arch/x86/boot/printf.o
CC      arch/x86/boot/regs.o
CC      arch/x86/boot/string.o
CC      arch/x86/boot/tty.o
CC      arch/x86/boot/video.o
CC      arch/x86/boot/video-mode.o
CC      arch/x86/boot/version.o
CC      arch/x86/boot/video-vga.o
CC      arch/x86/boot/video-vesa.o
CC      arch/x86/boot/video-bios.o
LD      arch/x86/boot/setup.elf
OBJCOPY arch/x86/boot/setup.bin
OBJCOPY arch/x86/boot/vmlinux.bin
BUILD  arch/x86/boot/bzImage
Kernel: arch/x86/boot/bzImage is ready (#1)
root@slack:/usr/src/linux-7.0.6#

```

Następnie należy zbudować moduły jądra, za pomocą polecenia `make modules`.

```

root@slack:/usr/src/linux-7.0.6# make modules
CALL    scripts/checksyscalls.sh
DESCEND objtool
INSTALL libsubcmd_headers
AS [M]  arch/x86/crypto/ghash-clmulni-intel_asm.o
CC [M]  arch/x86/crypto/ghash-clmulni-intel_glue.o
LD [M]  arch/x86/crypto/ghash-clmulni-intel.o

```

```
LD [M] drivers/gpu/drm/clients/drm_client_lib.o
CC [M] drivers/gpu/drm/ttm/ttm_tt.o
CC [M] drivers/gpu/drm/ttm/ttm_bo.o
CC [M] drivers/gpu/drm/ttm/ttm_bo_util.o
CC [M] drivers/gpu/drm/ttm/ttm_bo_vm.o
CC [M] drivers/gpu/drm/ttm/ttm_module.o
CC [M] drivers/gpu/drm/ttm/ttm_execbuf_util.o
CC [M] drivers/gpu/drm/ttm/ttm_range_manager.o
CC [M] drivers/gpu/drm/ttm/ttm_resource.o
CC [M] drivers/gpu/drm/ttm/ttm_pool.o
CC [M] drivers/gpu/drm/ttm/ttm_device.o
CC [M] drivers/gpu/drm/ttm/ttm_sys_manager.o
CC [M] drivers/gpu/drm/ttm/ttm_backup.o
CC [M] drivers/gpu/drm/ttm/ttm_agp_backend.o
LD [M] drivers/gpu/drm/ttm/ttm.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_execbuf.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_gmr.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_kms.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_drv.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_ioctl.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_resource.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_ttm_buffer.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_cmd.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_irq.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_ldu.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_overlay.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_gmrid_manager.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_fence.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_bo.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_scrn.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_context.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_surface.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_prime.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_mob.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_shader.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_cmdbuf_res.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_cmdbuf.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_std.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_cotable.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_so.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_binding.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_msg.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_simple_resource.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_va.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_blit.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_validation.o
CC [M] drivers/gpu/drm/vmwgfx/vmwgfx_page_dirty.o
```

```

LD [M] drivers/input/joydev.ko
CC [M] drivers/input/evdev.mod.o
LD [M] drivers/input/evdev.ko
CC [M] drivers/i2c/busses/i2c-piix4.mod.o
LD [M] drivers/i2c/busses/i2c-piix4.ko
CC [M] drivers/i2c/i2c-core.mod.o
LD [M] drivers/i2c/i2c-core.ko
CC [M] drivers/i2c/i2c-smbus.mod.o
LD [M] drivers/i2c/i2c-smbus.ko
CC [M] drivers/virt/uboxguest/uboxguest.mod.o
LD [M] drivers/virt/uboxguest/uboxguest.ko
CC [M] drivers/powercap/intel_rapl_common.mod.o
LD [M] drivers/powercap/intel_rapl_common.ko
CC [M] drivers/powercap/intel_rapl_msr.mod.o
LD [M] drivers/powercap/intel_rapl_msr.ko
CC [M] sound/soundcore.mod.o
LD [M] sound/soundcore.ko
CC [M] sound/core/snd.mod.o
LD [M] sound/core/snd.ko
CC [M] sound/core/snd-timer.mod.o
LD [M] sound/core/snd-timer.ko
CC [M] sound/core/snd-pcm.mod.o
LD [M] sound/core/snd-pcm.ko
CC [M] sound/pci/snd-intel8x0.mod.o
LD [M] sound/pci/snd-intel8x0.ko
CC [M] sound/pci/ac97/snd-ac97-codec.mod.o
LD [M] sound/pci/ac97/snd-ac97-codec.ko
CC [M] sound/ac97_bus.mod.o
LD [M] sound/ac97_bus.ko
CC [M] net/802/psnap.mod.o
LD [M] net/802/psnap.ko
CC [M] net/802/stp.mod.o
LD [M] net/802/stp.ko
CC [M] net/802/garp.mod.o
LD [M] net/802/garp.ko
CC [M] net/802/mrp.mod.o
LD [M] net/802/mrp.ko
CC [M] net/ipv6/ipv6.mod.o
LD [M] net/ipv6/ipv6.ko
CC [M] net/8021q/8021q.mod.o
LD [M] net/8021q/8021q.ko
CC [M] net/wireless/cfg80211.mod.o
LD [M] net/wireless/cfg80211.ko
CC [M] net/llc/llc.mod.o
LD [M] net/llc/llc.ko
CC [M] net/rfkill/rfkill.mod.o
LD [M] net/rfkill/rfkill.ko
root@slack:/usr/src/linux-7.0.6#

```

Kolejnym krokiem jest instalacja modułów, za pomocą komendy `make modules_install`.

```
root@slack:/usr/src/linux-7.0.6# make modules install
```

```

INSTALL /lib/modules/7.0.6/kernel/arch/x86/crypto/ghash-clmulni-intel.ko
INSTALL /lib/modules/7.0.6/kernel/fs/efivarfs/efivarfs.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/acpi/ac.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/acpi/button.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/acpi/battery.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/char/agp/agpgart.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/char/agp/intel-agp.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/char/agp/intel-gtt.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/gpu/drm/clients/drm_client_lib.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/gpu/drm/drm.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/gpu/drm/drm_ttm_helper.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/gpu/drm/drm_kms_helper.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/gpu/drm/ttm/ttm.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/gpu/drm/umwgfx/umwgfx.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/block/loop.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/net/ethernet/intel/e1000/e1000.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/usb/host/ehci-hcd.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/usb/host/ehci-pci.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/usb/host/ohci-hcd.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/usb/host/ohci-pci.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/input/serio/serio_raw.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/input/mouse/psmouse.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/input/joydev.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/input/evdev.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/i2c/busses/i2c-piix4.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/i2c/i2c-core.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/i2c/i2c-smbus.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/virt/vboxguest/vboxguest.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/powercap/intel_rapl_common.ko
INSTALL /lib/modules/7.0.6/kernel/drivers/powercap/intel_rapl_msr.ko
INSTALL /lib/modules/7.0.6/kernel/sound/soundcore.ko
INSTALL /lib/modules/7.0.6/kernel/sound/core/snd.ko
INSTALL /lib/modules/7.0.6/kernel/sound/core/snd-timer.ko
INSTALL /lib/modules/7.0.6/kernel/sound/core/snd-pcm.ko
INSTALL /lib/modules/7.0.6/kernel/sound/pci/snd-intel8x0.ko
INSTALL /lib/modules/7.0.6/kernel/sound/pci/ac97/snd-ac97-codec.ko
INSTALL /lib/modules/7.0.6/kernel/sound/ac97_bus.ko
INSTALL /lib/modules/7.0.6/kernel/net/802/psnap.ko
INSTALL /lib/modules/7.0.6/kernel/net/802/stp.ko
INSTALL /lib/modules/7.0.6/kernel/net/802/garp.ko
INSTALL /lib/modules/7.0.6/kernel/net/802/mrp.ko
INSTALL /lib/modules/7.0.6/kernel/net/ipv6/ipv6.ko
INSTALL /lib/modules/7.0.6/kernel/net/8021q/8021q.ko
INSTALL /lib/modules/7.0.6/kernel/net/wireless/cfg80211.ko
INSTALL /lib/modules/7.0.6/kernel/net/l1c/l1c.ko
INSTALL /lib/modules/7.0.6/kernel/net/rfkill/rfkill.ko
DEPMOD /lib/modules/7.0.6
root@slack:/usr/src/linux-7.0.6#

```

Potem należy zainstalować pliki nagłówkowe, za pomocą komendy `make headers_install`.

```

root@slack:/usr/src/linux-7.0.6# make headers_install
HOSTCC  scripts/unifdef
HDRINST usr/include/asm-generic/fcntl.h
HDRINST usr/include/asm-generic/mman-common.h
HDRINST usr/include/asm-generic/auxvec.h
HDRINST usr/include/asm-generic/int-l164.h
HDRINST usr/include/asm-generic/int-l64.h
HDRINST usr/include/asm-generic/ipcbuf.h
HDRINST usr/include/asm-generic/types.h
HDRINST usr/include/asm-generic/signal.h
HDRINST usr/include/asm-generic/termbits-common.h
HDRINST usr/include/asm-generic/ioctls.h
HDRINST usr/include/asm-generic/bpf_perf_event.h
HDRINST usr/include/asm-generic/stat.h
HDRINST usr/include/asm-generic/posix_types.h
HDRINST usr/include/asm-generic/msgbuf.h
HDRINST usr/include/asm-generic/errno-base.h
HDRINST usr/include/asm-generic/termbits.h
HDRINST usr/include/asm-generic/mman.h
HDRINST usr/include/asm-generic/hugetlb_encode.h

```

```
HDRINST usr/include/asm-generic/termios.h
HDRINST usr/include/asm-generic/signal-defs.h
HDRINST usr/include/asm-generic/siginfo.h
HDRINST usr/include/asm-generic/sockios.h
HDRINST usr/include/asm-generic/shmbuf.h
HDRINST usr/include/asm-generic/errno.h
HDRINST usr/include/asm-generic/param.h
HDRINST usr/include/asm-generic/unistd.h
HDRINST usr/include/cxl/features.h
HDRINST usr/include/drm/i915_drm.h
HDRINST usr/include/drm/drm_sarea.h
HDRINST usr/include/drm/amdgpu_drm.h
HDRINST usr/include/drm/uc4_drm.h
HDRINST usr/include/drm/tegra_drm.h
HDRINST usr/include/drm/armada_drm.h
HDRINST usr/include/drm/nouveau_drm.h
HDRINST usr/include/drm/panthor_drm.h
HDRINST usr/include/drm/omap_drm.h
HDRINST usr/include/drm/drm_mode.h
HDRINST usr/include/drm/rocket_accel.h
HDRINST usr/include/drm/msm_drm.h
HDRINST usr/include/drm/habanalabs_accel.h
HDRINST usr/include/drm/amdxdna_accel.h
HDRINST usr/include/drm/umwgfx_drm.h
HDRINST usr/include/drm/etnaviv_drm.h
HDRINST usr/include/drm/pvr_drm.h
HDRINST usr/include/drm/qxl_drm.h
HDRINST usr/include/drm/drm.h
HDRINST usr/include/drm/v3d_drm.h
HDRINST usr/include/drm/panfrost_drm.h
HDRINST usr/include/drm/drm_fourcc.h
HDRINST usr/include/drm/vgem_drm.h
HDRINST usr/include/drm/virtgpu_drm.h
HDRINST usr/include/drm/nova_drm.h
HDRINST usr/include/drm/asahi_drm.h
HDRINST usr/include/drm/radeon_drm.h
HDRINST usr/include/drm/ethosu_accel.h
HDRINST usr/include/drm/qaic_accel.h
HDRINST usr/include/drm/iopu_accel.h
HDRINST usr/include/drm/xe_drm.h
HDRINST usr/include/drm/exynos_drm.h
HDRINST usr/include/drm/lima_drm.h
HDRINST usr/include/fwctl/mlx5.h
HDRINST usr/include/fwctl/cxl.h
HDRINST usr/include/fwctl/pds.h
HDRINST usr/include/fwctl/fwctl.h
HDRINST usr/include/linux/mei.h
```



```

HDRINST usr/include/asm/mman.h
HDRINST usr/include/asm/msr.h
HDRINST usr/include/asm/setup_data.h
HDRINST usr/include/asm/sembuf.h
HDRINST usr/include/asm/ucontext.h
HDRINST usr/include/asm/processor-flags.h
HDRINST usr/include/asm/swab.h
HDRINST usr/include/asm/amd_hsmpt.h
HDRINST usr/include/asm/statfs.h
HDRINST usr/include/asm/ptrace-abi.h
HDRINST usr/include/asm/setup.h
HDRINST usr/include/asm/bootparam.h
HDRINST usr/include/asm/sgx.h
HDRINST usr/include/asm/prctl.h
HDRINST usr/include/asm/kvm_para.h
HDRINST usr/include/asm/ist.h
HDRINST usr/include/asm/bitsperlong.h
HDRINST usr/include/asm/boot.h
HDRINST usr/include/asm/mce.h
HDRINST usr/include/asm/siginfo.h
HDRINST usr/include/asm/svm.h
HDRINST usr/include/asm/shmbuf.h
HDRINST usr/include/asm/perf_regs.h
HDRINST usr/include/asm/posix_types_64.h
HDRINST usr/include/asm/vm86.h
HDRINST usr/include/asm/unistd.h
HDRINST usr/include/asm/usyscall.h
HDRINST usr/include/asm/vmx.h
HDRINST usr/include/asm/sigcontext32.h
HDRINST usr/include/asm/fcntl.h
HDRINST usr/include/asm/unistd_x32.h
HDRINST usr/include/asm/ipcbuf.h
HDRINST usr/include/asm/types.h
HDRINST usr/include/asm/ioctls.h
HDRINST usr/include/asm/bpf_perf_event.h
HDRINST usr/include/asm/termbits.h
HDRINST usr/include/asm/socket.h
HDRINST usr/include/asm/resource.h
HDRINST usr/include/asm/poll.h
HDRINST usr/include/asm/unistd_64.h
HDRINST usr/include/asm/ioctl.h
HDRINST usr/include/asm/termios.h
HDRINST usr/include/asm/unistd_32.h
HDRINST usr/include/asm/sockios.h
HDRINST usr/include/asm/errno.h
HDRINST usr/include/asm/param.h
INSTALL ./usr/include
root@slack:/usr/src/linux-7.0.6#

```

Kolejny krok to sprawdzenie zainstalowanych modułów w katalogu /lib/modules/7.0.6.

```

root@slack:/usr/src/linux-7.0.6# ls /lib/modules/7.0.6/
buidle      modules.alias.bin      modules.builtin.bin    modules.dep.bin        modules.softdep
kernel/     modules.builtin         modules.builtin.modinfo modules.devname         modules.symbols
modules.alias modules.builtin.alias.bin modules.dep             modules.order          modules.symbols.bin

```

Następnie należy skopiować pliki kernela do systemu do katalogu /boot, za pomocą polecenia: `cp arch/x86_64/boot/bzImage /boot/vmlinuz-custom2-7.0.6`.

```

root@slack:/usr/src/linux-7.0.6# cp arch/x86_64/boot/bzImage /boot/vmlinuz-custom2-7.0.6

```

Później trzeba przekopiować tablicę symboli używanych przez kernel, za pomocą komendy: `cp System.map /boot/System.map-custom2-7.0.6`.

```

root@slack:/usr/src/linux-7.0.6# cp System.map /boot/System.map-custom2-7.0.6

```

Potem należy przekopiować plik konfiguracyjny kernela, za pomocą polecenia:
`cp .config /boot/config-custom2-7.0.6.`

```
root@slack:/usr/src/linux-7.0.6# cp .config /boot/config-custom2-7.0.6
```

Następnie należy przejść do katalogu /boot, usunąć starą tablicę symboli i utworzyć link symboliczny dla tablicy symboli kernela.

```
root@slack:/usr/src/linux-7.0.6# cd /boot/
root@slack:/boot# rm System.map
root@slack:/boot# ln -s System.map-custom2-7.0.6 System.map
```

Kolejny krok to utworzenie dysku RAM, za pomocą wywołania skryptu i wykorzystania komendy, wygenerowanej przez ten skrypt.

```
root@slack:/boot# /usr/share/mkinitrd/mkinitrd_command_generator.sh -k 7.0.6
#
# mkinitrd_command_generator.sh revision 1.45
#
# This script will now make a recommendation about the command to use
# in case you require an initrd image to boot a kernel that does not
# have support for your storage or root filesystem built in
# (such as the Slackware 'generic' kernels').
# A suitable 'mkinitrd' command will be:
mkinitrd -c -k 7.0.6 -f ext4 -r /dev/sda1 -m ext4 -u -o /boot/initrd.gz
root@slack:/boot# mkinitrd -c -k 7.0.6 -f ext4 -r /dev/sda1 -m ext4 -u -o /boot/initrd-custom2-7.0.6.gz
49967 blocks
/boot/initrd-custom2-7.0.6.gz created.
Be sure to run lilo again if you use it.
root@slack:/boot#
```

Później trzeba ponownie skonfigurować GRUB.

```
root@slack:/boot# grub-mkconfig -o /boot/grub/grub.cfg
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-huge-5.15.19
Found initrd image: /boot/initrd.gz
Found linux image: /boot/vmlinuz-huge
Found initrd image: /boot/initrd.gz
Found linux image: /boot/vmlinuz-generic-5.15.19
Found initrd image: /boot/initrd.gz
Found linux image: /boot/vmlinuz-generic
Found initrd image: /boot/initrd.gz
Found linux image: /boot/vmlinuz-custom2-7.0.6
Found initrd image: /boot/initrd-custom2-7.0.6.gz
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
root@slack:/boot#
```

Następnie należy skopiować obraz jądra oraz ramdisku na partycję EFI.

```
root@slack:/boot# cp /boot/vmlinuz-custom2-7.0.6 /boot/efi/EFI/Slackware/
root@slack:/boot# cp /boot/initrd-custom2-7.0.6.gz /boot/efi/EFI/Slackware/
```

Kolejnym krokiem jest dodanie nowego wpisu do konfiguracji bootloadera ELILO.

```
root@slack:/boot# nano /boot/efi/EFI/Slackware/elilo.conf
```

```

GNU nano 6.0 /boot/efi/EFI/Slackware/elilo.conf
#chooser=simple
prompt
delay=15
timeout=30
default=custom2-kernel-7.0.6
#
image=vmlinuz
    label=vmlinuz
    initrd=initrd.gz
    read-only
    append="root=/dev/sda1 vga=normal ro"

image=vmlinuz-custom2-7.0.6
    label=custom2-kernel-7.0.6
    initrd=initrd-custom2-7.0.6.gz
    read-only
    append="root=/dev/sda1 vga=791 ro_"

```

Następnie należy zrestartować system, za pomocą polecenia reboot i w opcjach zaawansowanych wybrać odpowiedni kernel.

```

root@slack:/boot# reboot

Broadcast message from root@slack.localhost (tty1) (Sat May 16 17:08:34 2026):

The system is going down for reboot NOW!
INIT: Sending processes configured via /etc/inittab the TERM signal

Running shutdown script /etc/rc.d/rc.6:
INIT: version 3.01 reloading
Saving system time to the hardware clock (localtime).
Stopping haveged.
Unmounting remote filesystems:
Stopping system message bus...
Stopping the network interfaces...
Stopping udevd
Sending all processes the SIGTERM signal.
[ 7630.644566] elogind-daemon[1032]: Received signal 15 [TERM]

```

GNU GRUB version 2.06

Slackware-15.0 GNU/Linux
*Advanced options for Slackware-15.0 GNU/Linux
UEFI Firmware Settings

Use the ▲ and ▼ keys to select which entry is highlighted.
Press enter to boot the selected OS, 'e' to edit the commands before booting or 'c' for a command-line.

GNU GRUB version 2.06

Slackware-15.0 GNU/Linux, with Linux 5.15.19
Slackware-15.0 GNU/Linux, with Linux 5.15.19 (recovery mode)
Slackware-15.0 GNU/Linux, with Linux huge
Slackware-15.0 GNU/Linux, with Linux huge (recovery mode)
Slackware-15.0 GNU/Linux, with Linux 5.15.19
Slackware-15.0 GNU/Linux, with Linux 5.15.19 (recovery mode)
Slackware-15.0 GNU/Linux, with Linux generic
Slackware-15.0 GNU/Linux, with Linux generic (recovery mode)
*Slackware-15.0 GNU/Linux, with Linux custom2-7.0.6
Slackware-15.0 GNU/Linux, with Linux custom2-7.0.6 (recovery mode)

Use the ▲ and ▼ keys to select which entry is highlighted.
Press enter to boot the selected OS, 'e' to edit the commands before booting or 'c' for a command-line. ESC to return previous menu.

```

/dev/sda4 on /boot/efi type vfat (rw,relatime,fmask=0022,dmask=0022,codepage=437,iocharset=iso8859-1,shortname=mixed,utf8,errors=remount-ro)
Using /etc/random-seed to initialize /dev/urandom.
INIT: Entering runlevel: 3
Going multiuser...
Updating shared library links: /sbin/ldconfig &
Starting syslogd daemon: /usr/sbin/syslogd -s
Updating X font indexes: /usr/bin/fc-cache -f &
Updating hardware database index: /sbin/udevadm hudeb --update
Triggering udev events: /sbin/udevadm trigger --action=change
Starting haveged entropy daemon: /sbin/haveged
Starting the network interfaces...
eth0: polling for DHCP server
dhcpcd-9.4.1 starting
DUID 00:04:3b:3a:a8:d0:3b:2b:bb:4e:b2:67:99:51:2a:21:85:a4
eth0: waiting for carrier
eth0: carrier acquired
eth0: IFAID 27:88:e6:96
eth0: soliciting a DHCP lease
eth0: offered 10.0.2.15 from 10.0.2.2
eth0: probing address 10.0.2.15/24
eth0: leased 10.0.2.15 for 86400 seconds
eth0: adding route to 10.0.2.0/24
forked to background, child pid 780
eth0: adding default route via 10.0.2.2
Starting system message bus: /usr/bin/dbus-uuidgen --ensure ; /usr/bin/dbus-daemon --system
Starting elogind: /lib64/elogind/elogind --daemon
Starting OpenSSH SSH daemon: /usr/sbin/sshd
Starting ACPI daemon: /usr/sbin/acpid
Updating MIME database: /usr/bin/update-mime-database /usr/share/mime &
Updating gtk.immodules:
  /usr/bin/update-gtk-immodules &
Updating gdk-pixbuf.loaders:
  /usr/bin/update-gdk-pixbuf-loaders &
Compiling GSettings XML schema files:
  /usr/bin/glib-compile-schemas /usr/share/glib-2.0/schemas &
Starting crond: /usr/sbin/crond -l notice
Starting atd: /usr/sbin/atd -b 15 -l 1
Loading /usr/share/kbd/keymaps/i386/german/qwertz.pl.map.gz
Starting gpm: /usr/sbin/gpm -m /dev/mouse -t imps2

Welcome to Linux 7.0.6 x86_64 (tty1)

slack login: root
Password:
Last login: Sun May 17 16:46:17 on tty1
Linux 7.0.6.
root@slack:~# uname -r
7.0.6
root@slack:~#

```

Wnioski:

Udało się skompilować kernel, za pomocą obu metod, pomimo jednego błędu podczas kompilacji za pomocą metody localmodconfig, który na szczęście udało się łatwo naprawić. Kompilacja tym sposobem przebiegła dosyć sprawnie i była przyjemna w jej realizacji. Natomiast kompilacja streamlineconfig.pl wymagała więcej czasu. Generalnie kompilacja, za pomocą localmodconfig bardziej przypadła mi do gustu.